Assignment-Aligned Effective States in a 2-SAT Transformer

Protected label tests, learned-basis controls and exact readout accounting

Anonymous

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Abstract

Cammarata's *Neural Networks Are Already Clean* proposes that native neurons can be structured building blocks even when broader representations look distributed. We test one published interpretation of a five-variable 2-SAT transformer. On a protected 7,252-formula bank, seventeen source-nominated assignment labels rank strongly (macro area under the receiver operating characteristic curve (AUROC) 0.991), but the frozen combined and threshold-fidelity criteria fail. A separately protected complete-layer comparison places native assignment matching above 63 fixed full-space rotations; their different sparsity and sign geometry prevent an exchangeability interpretation. Retrospective analysis finds that 34 of the 512 units contain essentially all observed post-rectified-linear-unit (ReLU) energy and form a numerical-rank-34 state. Native assignment matching exceeds fixed rotations and positive mixtures within that state. A locally fitted nonnegative matrix factorisation slightly improves matching (0.986 versus 0.984), recovers 32 native axes at cosine at least 0.99, and incurs 3.15% reconstruction error despite an available exact identity factorisation. The final readout sums graded positive SAT evidence. Fixed-readout threshold substitution gives 98.15% task accuracy; separately fitted state-only truth classifiers give 99.12% from bits and 99.64% from continuous values. An exact clause-edit collision shows that neither the bit report nor the complete current truth table determines every semantic edit outcome. This single checkpoint is compatible with both Already Clean and superposition accounts. The contribution is a reproducible case study separating semantic ranking, threshold fidelity, effective-state geometry, final-cut dependence and action sufficiency, with explicit limits on their use in alignment research.

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1 Introduction

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A neuron interpretation can answer one question well and still be inadequate for another. A Boolean label may describe which assignment a SAT-model unit responds to, while discarding information needed to predict what happens when a clause is replaced. A labelled signal may also rank cases correctly and affect the decision while falling below the source's binarisation threshold. These are different limitations and require different tests.

The immediate context is Cammarata's informal research essay *Neural Networks Are Already Clean*. Its working hypothesis is that a network's native neurons and circuits are often legible building blocks, and that famous examples of apparently arbitrary polysemanticity were misread. The proposed alternative is **recursive generalisation**: a feature grows related branches across contexts, so a neuron may join cats, vehicles or linguistic situations through a shared relation rather than an arbitrary capacity-saving code. The essay also attributes apparent messiness to architectural bottlenecks and imperfect interpretability tools, argues for studying native mechanisms directly rather than treating a learned decomposition as a necessary first step, and presents broad preliminary prevalence estimates. These are several claims, not one binary thesis (Cammarata, source snapshot).

This paper does not attempt a census of clean neurons or adjudicate whether recursive generalisation explains most polysemanticity. It asks an operational follow-up: **what additional uses does one published native-neuron interpretation support?** We study an independently published interpretation of a small transformer trained on five-variable, ten-clause 2-SAT (Palumbo et al., 2025). The source validates component and prefix approximations of the model's current computation. It does not claim that its final 34-bit report is a sufficient state for every future input-edit query. A useful result may therefore be positive, negative or a diagnosis that the proposed target asks the report to retain information outside its original semantic object. None is, by itself, a refutation of the source study or of Cammarata's broader hypothesis.

The task itself supplies a strong alternative account of this structure. Five-variable SAT can be expressed as an OR over 32 assignments, each satisfying a conjunction of clause-absence conditions. A graded nonnegative code related to those assignments, followed by a positive readout, is therefore compatible with the task's logical structure and the source's weighted-count interpretation. This is an explanatory alternative, not an identified training cause.

The increment over Palumbo et al. is an audit of that interpretation, not a new discovery of its basic mechanism:

Reproduced or retained from the source	Newly quantified on the declared banks	Newly tested uses and comparisons
Effect-based selection of 34 units; approximate logical rules and source threshold 0.5; positive weighted-count account	Numerical effective width, assignment-ranking and threshold-error localisation, graded margins, and exact final-cut ablation accounting	Protected all-seventeen semantic stress test; protected full-layer matching and retrospective geometry-matched, probe and learned-basis controls
A report describing the model's current computation, without a universal edit-sufficiency guarantee	Exact finite-bank ambiguity of the report and current truth table for Boolean clause-edit outcomes	A concrete report/action collision, fixed action-selection learners and named-unit response forecasts, each with its own target

The source already permits imperfect thresholds and distribution-dependent approximation. Our adverse results concern the specified new criteria or uses, not a contradiction of an exact source guarantee.

Three levels organise the evidence. **Legibility** asks whether a unit responds in accordance with a proposed semantic label, including after a symmetry such as variable renaming. **Operational sufficiency** asks whether the resulting report retains enough information, under a declared decoder and context, to answer a new response or action question. **Causal mechanism** asks whether an intervention follows the natural computational route rather than merely changing a downstream logit. A result at one level does not settle the others.

The paper presents the protected label study, a separate response-blind source-grammar audit, complete-layer analysis, exact readout accounting and one edit-response witness. A separate Supplement retains Appendices A–L, including the mathematical calibrations, source audits, earlier pilots and new-bank provenance.

The point of contact with *Already Clean* is bounded but direct. The final multilayer perceptron (MLP) has nominal width 512 for a task with 32 assignments and 40 clause types. Palumbo et al. selected 34 units whose largest direct effect on the SAT output exceeded 0.1, then derived logical rules for those units; seventeen rules name one satisfying assignment. This is a source-nominated causal report, not a random neuron sample. Complete-layer extraction changes the capacity description: on the audited bank the state has effective rank 34 despite the nominal width. Whether weight decay, task structure or another training contingency caused that economy is unidentified. This regime is compatible with Cammarata's emphasis on native structure, but it is also compatible with superposition accounts under suitable feature and capacity assumptions; nominal width 512 and effective rank 34 for 32 assignment labels do not establish how many features the model represents, and the checkpoint cannot discriminate those explanations. Polysemanticity can depend on feature capacity (Scherlis et al., 2022) and can also arise incidentally (Lecomte et al., 2024). We test whether the fixed singleton labels rank fresh cases, survive the source threshold, support specified new uses, and sit inside a broader assignment report when the complete layer is inspected. A fixed-rotation comparison is retained as one protected diagnostic, but rotating a sparse nonnegative ReLU state

makes dense signed coordinates, so native and rotated bases are not exchangeable under a generic null. Learned decompositions are therefore evaluated retrospectively. The study does not adjudicate recursive development, prevalence or the causal origin of the effective-width collapse.

Proposal or question	What is tested here	Finding and scope
Published singleton-assignment labels rank cases and support faithful source-threshold events	All seventeen source-nominated units in one epoch-900 checkpoint; five semantic strata per assignment, balanced near/far SAT negatives, plus dependent variable-transposition checks	All three pre-specified ranking gates pass (assignment-macro AUROC 0.991; unit range 0.961–1.000). A threshold-specificity guard fails for two units and is shared with the fidelity criterion; four units also fail at least one positive fidelity cell
The effective final-MLP state is strongly assignment-aligned	Complete 512-unit extraction; protected matching against 63 fixed full-space rotations; retrospective rank, purity, learned-basis, prevalence and linear-probe controls	The source-nominated 34 units form the bank-effective numerical-rank-34 state. Native matching exceeds fixed rotations and mixtures, but learned 34-component nonnegative matrix factorisation (NMF) slightly exceeds its mean while recovering 32 native axes. This is one checkpoint and engineered bank, with source selection fully disclosed
A compact neuron report can guide a new intervention choice	Palumbo et al.'s 34-bit report on legal clause replacements	It guides choices far above action context alone but is not universally sufficient; the exact collision shows that current truth is insufficient for the target, while truth-table refinement also removes report-specific lookup errors
The live report contributes at the exact final affine readout	Exact individual and grouped zeroing; per-threshold-category margin accounting; within-stratum resampling; earlier four-unit origin-restoration pilot	Every live report unit has a positive SAT-minus-UNSAT coefficient. Joint removal crosses more argmax margins than individual removals because the logit contributions are additive and redundant. Earlier restoration changes 115/126 initially correct edited decisions. These final-cut interventions do not identify an upstream route
Most neurons in modern networks are clean, or recursive generalisation explains most apparent mixtures	Not tested by this single-checkpoint audit	Outside scope; the paper records current direct follow-ons without pooling them into its own evidence

The execution-status table below is the principal chronology record. The empirical presentation starts with the all-unit label study, then the full-layer comparison and readout, followed by the edit-response audit. Protected criteria remain unchanged by later diagnostics; the all-unit combined and fidelity decisions share a failed guard and are not independent. Supplement Appendix J retains the detailed execution and replay lineage.

Here **protected** means that the finite inputs, selection rule, primary criteria and analysis code were fixed internally before the checkpoint responses were generated, and later diagnostics cannot change the recorded primary decision. The hash-chained records establish artifact identity; they are not an external timestamp or independent registration.

Term	Meaning in this paper
Protected / pre-specified	Inputs, criteria and analysis were locally frozen before the relevant new checkpoint responses; prior work on the same units and checkpoint still limits confirmatory force
Opened	Outcomes from the bank had already been inspected before the stated diagnostic was designed or fitted
Report	The declared information retained from a neural state, such as the 34 threshold bits
Contract	The response target, permitted report and context, decoder class, tolerance and cost ledger evaluated together
Readiness	A post-action threshold event predicted for one named unit from a pre-action unit score and edit semantics

Execution and presentation use different orders. The table below gives execution order; Section 4 presents the all-unit result first because it is the central label audit.

Execution stage	Evidence status	Primary question
Edit lookup and first action-selection pilot	Opened development bank	Does the 34-bit report retain enough information to predict or choose clause edits?
Negative-score task-policy screen	Frozen development contract on cached opened outcomes	Does one extra sub-threshold bit improve edit policy performance?
Readiness study	Frozen development contract on cached opened states	Can a negative score plus edit semantics forecast a named unit's threshold event?
Four-direction matched-edit pilot	Protected fresh bank	Do four selected labels transport across consequential edits and one variable swap?
All-seventeen label study	Protected fresh bank	How do all published singleton labels rank, threshold and generalise across five semantic strata?

Execution stage	Evidence status	Primary question
Full-layer sparse-report and readout study	Protected re-extraction of the same complete bank, followed by labelled post hoc localisation	How is assignment alignment distributed across all 512 coordinates, where do the protected matches lie relative to the source report, and which units affect the exact final readout?
Versioned diagnostics	Post hoc, zero-forward	Where do errors concentrate, and which alternative descriptors remain confounded?

The audit distinguishes the response question, permitted report and context, decoder, tolerance and resource costs. Information loss, decoder expressivity, fitting error and failure to transfer are separate quantities. In the SAT case we apply the unrestricted finite-bank information audit exactly, while the fixed gradient-boosting screens are learner diagnostics and do not estimate the optimum of a certified decoder class. Supporting digit-model experiments make the restricted-decoder distinction exactly checkable: sparse rational contrasts reject every affine decoder even when reports are distinct; the full tested quartile library remains inadequate; a quantised native-state reference meets the bound; and a target-dependent encoder is much shorter once its computation is charged. These calibrations use established optimisation and state-abstraction tools. They are supporting evidence for the audit, not the mechanism behind the SAT findings.

The experiments preserve the essay's strongest interpretive latitude: a feature may be relational, conditional, signed or shared across object categories. Semantic cleanliness does not require one object noun per neuron. Our tests target declared predictions and uses rather than penalising a label for being richer than a noun.

The paper therefore makes a focused empirical contribution: a reproducible account of which additional uses of a published neuron interpretation succeed, fail or remain unresolved, with explicit negative results and costs. Exact decoder certificates, digit calibrations, source-aggregate audits and Inception provenance are retained in self-contained appendices. Their roles are stated separately rather than pooled into evidence for a single newly discovered mechanism.

The route through the paper is as follows. The next section relates the question to Cammarata's essay and to prior validation, probing and causal-abstraction work. The audit contract then separates information loss from decoder and fitting limitations. The central SAT section presents the label audit, sparse-report localisation, readout effects and a claim-by-claim ledger. Later sections keep the exact calibration and source audits separate, extract a reusable workflow, state implications for alignment research, and record limitations and reproducibility boundaries. The appendices contain the complete proofs, controls and provenance needed to assess the narrower claims.

2 Relation to "Already Clean" and prior validation

Native reports, probes and new uses. Cammarata's proposal is partly a response to superposition and dictionary-learning accounts in which models encode more features than available dimensions and sparse autoencoders recover more nearly monosemantic coordinates (Elhage et al., 2022; Bricken et al., 2023). Our protected full-layer experiment supplies one bounded native-versus-random-basis comparison; it does not adjudicate the general basis-level dispute. The wider study asks whether one native report supports specified new uses. Palumbo et al. (2025) validate component and prefix interpretations of a SAT transformer; our later experiment asks a different, prospective input-edit question. Neural SAT work establishes that learned systems can acquire useful satisfiability

structure, from NeuroSAT's built-in permutation symmetry and single-bit supervision to trained and constructed transformer solvers (Selsam et al., 2019; Pan et al., 2025). Those results concern solving ability rather than the sufficiency of one internal interpretation for a new edit policy. Hase et al. (2023) provide the closest direct antecedent: causal-tracing localisation does not identify the best layer for factual editing. Causal scrubbing tests whether an interpretation remains sufficient under resampling interventions (Chan et al., 2022), while the IOI circuit work separates faithfulness, completeness and minimality (Wang et al., 2023). Probe controls and counterfactual removal likewise distinguish recoverable information from information used by the task (Hewitt and Liang, 2019; Ravichander et al., 2021; Elazar et al., 2021).

Capacity pressure is only one proposed source of polysemanticity. Scherlis et al. (2022) analyse feature capacity and show in a toy model that importance, sparsity, kurtosis and architecture affect which features receive monosemantic or shared capacity. Lecomte et al. (2024) show theoretically and experimentally that random initialisation, regularisation and noise can also produce incidental mixed selectivity. Small algorithmic networks can exhibit unusually clean mechanisms after weight-decay-driven cleanup (Nanda et al., 2023), yet related tasks also show arbitrary circuit details across seeds (Chughtai et al., 2023) and qualitatively different algorithms under small hyperparameter or initialisation changes (Zhong et al., 2023). Dead or bank-silent neurons are also common enough that nominal width is an unreliable capacity measure (Voita et al., 2023). The SAT checkpoint is therefore informative as one precisely audited regime, not as a clean-versus-superposition population experiment. We execute one-to-one assignment matching across all 512 native coordinates, then interpret it using the bank-effective state, dimension-matched rotations, standardised rotations, positive mixtures and basis-invariant probes. Every comparison remains limited by the sparse nonnegative geometry, source selection, one checkpoint and one engineered bank.

Individual-neuron semantics also have a substantial prehistory. Network Dissection scores alignment between hidden units and a broad visual-concept vocabulary and finds that axis alignment can carry interpretable structure beyond discriminative performance (Bau et al., 2017). Its comparison of native units with random orthogonal rotations is the direct methodological precedent for our Haar reference ensemble: rotations preserve the network function after the compensating inverse transformation while changing coordinate-level interpretability. Elhage, Lasenby, and Olah (2023) show that even transformer residual streams can acquire a privileged coordinate basis, while also distinguishing that empirical fact from architectural invariance. A post-ReLU MLP state has a still stronger architectural basis asymmetry because nonnegativity and sparsity are destroyed by a generic rotation. Mu and Andreas (2020) replace atomic descriptions with logical compositions and find both coherent abstract features and unrelated mixtures. Gurnee et al. (2023) use sparse probes over more than 100 features in seven language models, including single-neuron localisation, and report layer- and scale-dependent sparsity. These methods establish ways to discover and score candidate neuron meanings. The present audit starts later in the pipeline: it takes one independently published label set as fixed and asks whether the scores, source threshold and permitted report support additional claims.

Ranking a neuron for a property is also distinct from showing that the model uses it. Antverg and Belinkov (2022) identify this as a central pitfall in individual-neuron analysis, and Sajjad, Durrani, and Dalvi (2022) survey neuron discovery, evaluation and use. Cross-seed consistency supplies another axis: Gurnee et al. (2024) find a minority of activation-matched universal neurons across five GPT-2 seeds and analyse their functional roles. These precedents motivate keeping the one-checkpoint ranking result separate from the direct-readout pilot and from any claim of a stable mechanism.

Neuron explanations require both sensitivity and specificity. Automated simulation scoring tests whether a proposed verbal description predicts activations (Bills et al., 2023). Huang et al. (2023) separate observational coverage from causal mediation and find that confident natural-language explanations can still have substantial errors and little causal efficacy. Bolukbasi et al. (2021) show that top-activating examples can support incompatible interpretations across datasets. These

precedents motivate separating sole- and multi-solution positives from UNSAT, Hamming-one and farther satisfiable negatives, and treating a threshold miss differently from absence of a ranked signal. AUROC evaluates ordering across the constructed positive and negative sets; it does not validate the source's chosen 0.5 event. We call the latter **threshold fidelity**, reserving probability calibration for the separate question of whether reported confidence matches empirical correctness frequency (Guo et al., 2017).

The report/decoder distinction has direct formal ancestors. Predictive $\mathcal{V}$-information makes informativeness relative to an observer class (Xu et al., 2020); conditional probing measures usable information beyond a declared baseline (Hewitt et al., 2021); and information-theoretic probing separates information in a representation from the convenience of one simple probe (Pimentel et al., 2020). Our finite e_{info} and $e_{\mathcal{G}}$ quantities are task-specific worst-case analogues rather than new general information measures. The contribution is a bounded trained-system audit with exact report/action interfaces, not the general observation that localisation, prediction and editing differ.

Palumbo et al.'s weighted-count account already motivates the importance of margins. Negative preactivation information and inactive-feature potential also have precedents in activation-boundary distillation, circuit tracing, SASFT and counterfactual activation potential (Heo et al., 2019; Ameisen et al., 2025; Deng et al., 2026; Swain and Dutta, 2026). Makelov et al. (2024) show that successful subspace patching can give an illusory semantic attribution, and Elhage et al. (2021) supply the additive direct-logit framework relevant to our final clamp. Narang's (2026) study of the same Inception channel is direct prior art for quantitative channel decomposition; our calibration does not reproduce that study or decide its semantic interpretation.

Rapid public tests of the motivating essay. Public work appeared within days of the essay. Our search through 18 October 2026 found two repositories, each self-described as preregistered, that study the canonical Inception channel and report natural-content structure together with limited abstract-shape transfer and category-sensitive input routes (Long, 2026; Eiras, 2026). Other pinned repositories examine recursive-development claims, weak-activation ablation, native neurons against alternative bases, label-to-steering agreement and controlled-policy sufficiency across language and vision models (Bhatt, 2026; Yandell, 2026; Phoenix, 2026; Li, 2026; Mittal, 2026; Boix, 2026; `whp199`, 2026; DongerBot, 2026). These artifacts are unrefereed, only days old and heterogeneous. We inspected pinned code or saved outputs where available but reran no neural forwards. They establish topical context and open questions; they are not pooled into evidence for this paper's SAT claims. The archived search ledger retains exact commits and extracted values. No directly responsive arXiv paper was identified by that cutoff; the new arXiv work cited below addresses adjacent questions of identifiability, composition, native readout and intervention reliability.

Identifiability and cleanliness. Zhou and Klindt (2026) prove that, under their supervised latent-variable assumptions and sufficient classifier-visible capacity, every population cross-entropy optimum is linear in the latents up to a readout-nullspace term. Under stronger generative assumptions—exact capacity, non-negative latent support and activations, fixed within-class variance, and minimum energy—they obtain axis alignment up to permutation and positive scaling; under sparse discriminative assumptions they instead obtain compressed-sensing recovery. These results clarify when clean coordinates can be statistically identifiable. Their general result does not make the classifier-visible projection sufficient for a new action menu: the nullspace term is unconstrained by classification loss, and intervention-policy sufficiency is indexed by the target, context and action set. In their exact-capacity axis-aligned case, however, the complete representation is invertible in the latent state and is therefore information-sufficient for any deterministic target fully determined by that state and declared context. This still supplies neither a low-cost decoder nor evidence that a real trained system satisfies the theorem's data-generating and optimality assumptions. Their signed-support ablation marks a relevant boundary for non-negativity, but negative network preactivations are not thereby identified with their model's ground-truth latents.

Compositionality and native readers. Ma, Mei, and Sala (2026) distinguish linear activation reconstruction from behavioral composition. Across 18 model-and-behavior settings, learned SAE

coordinates have lower first-order residuals than sum-preserving random rotations of the same selected subspace, yet pairwise terms improve held-out coalition ordering in every setting; in combined steering, pairwise modelling raises rank fidelity from 0.5874 to 0.7790. Thus coordinates may be comparatively clean one at a time while their joint downstream effects require interactions. Gu et al. (2026) make a complementary point about readout: intermediate residual states are most faithfully interpreted in a context-dependent frame induced by the remaining model, rather than by one fixed lens per layer. Both results support target-specific evaluation. They do not show that any particular neuron report fails, but they rule out the shortcut from individually readable coordinates to an additive, action-sufficient joint description.

Causal abstraction, symmetry and intervention methodology. Interchange interventions experimentally test whether aligned neural variables realise the causal properties of a high-level model (Geiger et al., 2022), while the later causal-abstraction framework formalises intervention transformations and graded faithfulness (Geiger et al., 2025). Distributed alignment search extends this programme beyond disjoint neuron sets (Geiger et al., 2023). Naturally occurring equivariance gives a direct precedent for asking whether transforming an input exchanges transformed neuron features (Olah et al., 2020). Activation-patching results depend materially on corruption, metric and patch design, so a zeroing or restoration result must be indexed by its exact intervention (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024). Predicting a selected response bank is weaker than validating a complete high-level causal model. The same target dependence is classical in state abstraction and predictive-state representations: what may be discarded depends on the planning or action-conditional prediction problem (Li, Walsh, and Littman, 2006; Littman, Sutton, and Singh, 2002). Sutter et al. (2025) show how sufficiently permissive alignment maps can trivialise abstraction under their assumptions. Our explicit decoder class addresses a restricted instance of that problem. It does not justify affine decoding universally. Méloux et al. (2026) evaluate causal-abstraction metrics on simulated systems and introduce Causal Abstraction Error. Their object is the faithfulness of a model/map triple; ours is feasibility and repair within a declared report/decoder class. The two evaluations can be used together.

Interpretability evaluation. RAVEL tests controlled attribute changes and preservation of other attributes (Huang et al., 2024). SAEBench evaluates multiple practical and representational criteria (Karvonen et al., 2025). Native compositional neuron groups provide a substantive alternative to a single-neuron/SAE dichotomy (Shafran et al., 2026). In a direct unrefereed response to the motivating essay, Gao (2026) argues that matching activating examples tests only one direction of a proposed label and advocates substituting simulated activations while scoring downstream KL. That proposal and our report-sufficiency audit share a falsification principle, but differ in object: his suggested test evaluates a neuron's predicted scalar over a broad corpus, whereas our central study evaluates a fixed multineuron report for declared edit-response and action targets. Zavattari, Tommasi, and Prencipe (2026) prospectively separate causal localisation, probe-gated steering and weight editing across five eligible model families; their fixed geometric regime predictor fails on two out-of-family tests and is retained without replacement. That stopping practice is relevant to our separation of targets and protected negative results, although their agent-failure intervention is a different task. Benchmark-reliability work motivates known-answer controls, repeated seeds, and comparisons that distinguish a metric from its intended construct (Chanin, 2026). Probe description-length work already charges aspects of decoder effort (Voita and Titov, 2020). Our bit and coefficient ledger is a transparent resource accounting, not a replacement for those measures or a universal measure of interpretability.

3 The audit contract

Let a deterministic downstream computation be $F(H, C)$, where H is the nominated activation state and C includes every permitted bypass or side input. For a fixed-side-state intervention, the target is

$$Y = F(H', C) - F(H, C).$$

If the intervention also changes C, both side states must be specified. A deterministic representation derived from the complete H cannot create information missing from H. The identity report therefore provides a necessary calibration: an unrestricted decoder that simply reruns the network proves no simplification.

An audit fixes the checkpoint, numerical interpretation, admissible states and interventions, output scale, report encoder T, allowed context K, decoder class $\mathcal{G}$, error tolerance ε, and cost ledger. Here K denotes exactly the variables actually supplied to the decoder; it need not contain all physical side state C. A paired report is $(T(H), T(H'), K)$. A decoder may intentionally use only the difference $T(H') - T(H)$; that additional restriction can lose information and must be recorded. We use three different quantities:

$$e_{\text{info}}(T; K) = \inf_{g} \max_{i} \|Y_i - g(T(H_i), T(H_i'), K_i)\|_\infty,$$

$$e_{\mathcal{G}}(T; K) = \inf_{g \in \mathcal{G}} \max_{i} \|Y_i - g(T(H_i), T(H_i'), K_i)\|_\infty, \qquad e_{\text{fit}}(T; K) = \max_{i} \|Y_i - \widehat{g}_i\|_\infty.$$

On a common finite bank, $e_{\text{info}} \le e_{\mathcal{G}}$. The further inequality $e_{\mathcal{G}} \le e_{\text{fit}}$ holds when the fitted decoder $\widehat{g}$ belongs to the declared class $\mathcal{G}$; otherwise e_{fit} is a separate fitted-model quantity. These are different from held-out risk. The finite unrestricted oracle is allowed to memorise: its success is not generalisation evidence. In the digit calibration of Appendix B, squared-error summaries sum over all ten target coordinates before averaging over cases.

The central SAT edit audit uses a different loss and interface. Its target is Boolean-enumerated post-edit UNSAT, not the transformer's post-edit argmax. Its unrestricted finite-bank statistic is the minimum empirical 0–1 lookup error within classes having identical $(T(\varphi), a)$. Native forwards separately check the four collision-witness endpoints and the specified neural-response cohorts; they do not supply the targets for all 399,360 assessment edits. A conflicting class has worst-case absolute error $1/2$ regardless of its minority count, so the SAT majority-vote floor is not e_{info} under the maximum-error definition above. We report both quantities explicitly and reserve the $e_{\text{info}}/e_{\mathcal{G}}$ notation for the paired continuous calibrations to which it applies.

For an unrestricted decoder under maximum error, the exact information error is half the largest coordinate range within cases having identical complete decoder inputs, including K. Supplying a varying action identifier can split these classes; its effect cannot be assessed while silently omitting it from the formal input. A colour-class version recovers the characteristic-graph formulation when the encoder itself is unconstrained. Neither result supplies a low-cost executable explanation. For the exact calibration in Appendix A, design matrix A instead contains the fixed decoder features. An intercept is included only where explicitly allowed.

This paired criterion is retrospective: computing $T(H')$ may require executing the intervened network. A prospective explanation instead predicts from $(T(H), a, K)$, where a is the declared intervention. These are different contracts. In particular, setting $T(H) = F(H, C)$ can make retrospective decoding trivial when all required side state is available to the encoder. Passing the paired audit alone therefore establishes neither a state-transition model nor a causal mechanism.

Identifiability-to-action remark. Let $h(z) = Az + \nu(z)$ and $U\nu(z) = 0$ as in Zhou and Klindt's classifier-optimal representation, and let $V(z) = P_U h(z)$ be its readout-visible projection. For a deterministic intervention target $Q(z, a, K)$, an unrestricted policy based on $(V(z), a, K)$ exists exactly when Q is constant on every fibre of that complete input. Classification optimality guarantees this factorisation for the trained class posterior; it gives no such guarantee for an arbitrary Q. A pair with the same (V, a, K) and different Q is therefore a complete finite obstruction. Conversely,

when the full report is invertible in z and Q is determined by (z, a, K), the information obstruction vanishes, although decoder complexity, numerical robustness and out-of-sample validity remain separate. This is the ordinary factorisation criterion applied to the identifiability result, not a new identifiability theorem.

The resource ledger separates state-report bits; number and precision of encoder weights and thresholds; decoder coefficients and permitted interactions; supervision used to discover features; and information supplied as context. A single real coordinate with unbounded precision and a lookup decoder is not an economical explanation merely because its coordinate count is one.

The contracts used below have distinct targets and information interfaces:

Contract	Available query inputs	Required result	Scope of a certificate
Paired response audit	Both endpoint reports and declared context	Output difference within tolerance	Fixed bank and declared decoder class
Prospective response	Initial report, action and context	Post-action output or its difference	Stated state/action domain; no native endpoint report
Exact linear state closure	Linear initial report and action word	Exact outputs and report updates	Linear identities on the specified reachable span
Approximate state propagation	Quantised initial state and exact action word	Outputs within the propagated bound	Allowed attenuations, without repeated requantisation
Action choice	Initial report, class context and fixed menu	Best margin reduction or a class-changing action	Finite decision problem; does not require response reconstruction

4 Central case: reusing a published SAT interpretation

Our central case is an independently published neuron interpretation. Palumbo et al. (2025) interpret a transformer trained on five-variable, ten-clause 2-SAT. Their final abstraction thresholds 34 nominated second-block MLP units at 0.5; seventeen of those units have a general rule naming one satisfying assignment. Their component/prefix validation does not claim that the final report can answer new input-edit queries, that each threshold is uniformly faithful on every semantic stratum, that assignment alignment is unusually concentrated in the native basis, or that the nominated units form a complete causal state. We therefore test four distinct objects: report sufficiency for an edit target, prospective prediction of named-unit responses, label discrimination for all seventeen singleton-assignment units, and full-layer assignment alignment relative to fixed random rotations. Direct-readout interventions are kept separate from natural-route causal claims.

We execute the authors' epoch-900 checkpoint, with its original model computation: two transformer blocks, one then four attention heads, width 128, ReLU MLP width 512 and 403,840 parameters. Each clause uses two distinct variables, printed in increasing index order with independent signs. There are 40 clause types. Our engineered banks allow repeated clauses; the source RandomCNFs generator instead samples ten distinct clauses without replacement. Inputs concatenate ten four-token clauses and a final readout token. We generate 4,096 distinct initially SAT formulas by independent uniform clause sampling conditional on satisfiability, using seed 2026101511. Entire origins split into 3,072 fit and 1,024 assessment cases. This conditional development bank differs from the source's balanced training/test distribution. The initial prediction studies below reuse this opened bank and perform no retraining of the checkpoint. The two later label studies use separately

frozen banks of fresh queries; the final one draws from the complete native 40-clause grammar and covers all seventeen singleton-assignment units.

Write $s_i(\varphi)$ for unit i's final-position preactivation and $b_i(\varphi) = \mathbf{1}[\max(s_i(\varphi), 0) \geq 0.5]$. The published report T_{pub} retains b_i at

$$J = (10, 29, 36, 41, 48, 55, 77, 81, 96, 150, 185, 189, 195, 201, 224, 231, 245,\\ 261, 291, 304, 317, 326, 334, 374, 380, 411, 416, 435, 450, 482, 490, 492, 495, 499).$$

A query supplies $(T_{\text{pub}}(\varphi), a)$, where $a = (s, c, d)$ replaces the actual clause c in slot s by $d \neq c$. Each decision considers the 39 replacements for one stated slot, without combining all ten menus and thereby revealing the formula. The action is known to the decoder. Boolean enumeration supplies the semantic edit targets. Where a native post-edit response is measured, the complete transformer is rerun; those measurements are not fixed-side-state hidden interventions and have their own stated cohorts.

4.1 Protected all-unit test of the published assignment labels

This section is the paper's pre-specified test of label validity. The motivating proposal, the tested object and the finding are deliberately separate. **Cammarata's proposal** is that native units and circuits are often structured, legible building blocks and that some apparent polysemanticity reflects recursive generalisation rather than arbitrary superposition. **The tested object** is much narrower: the seventeen final-block MLP units in one epoch-900 Palumbo et al. checkpoint whose published general rule names a single five-bit satisfying assignment. We measure each unit's last-token preactivation and the source's threshold event $s_u \geq 0.5$ on a fresh engineered formula bank. **The finding** is strong graded label ranking with heterogeneous threshold fidelity. All three threshold-free ranking gates pass. The frozen combined criterion is not met only because its fourth component is a source-threshold specificity guard; that same guard also appears inside the source-threshold fidelity criterion, which additionally fails four positive cells. The two negative protocol decisions are therefore not independent. This result neither estimates how common clean units are nor tests their developmental origin or natural causal route.

4.1.1 Protected design and criteria

The frozen design covers sixteen distinct assignments because units 55 and 416 share the label TTTFF. For each assignment it selects 32 formulas in each of five strata: the assignment is the sole solution; it is one of several solutions; the formula is UNSAT; the formula has one different solution; or it has several solutions excluding the assignment. Each satisfiable-negative stratum is split into sixteen formulas whose closest solution is one bit flip from the target and sixteen whose closest solution is at least two flips away. The bank therefore tests both sensitivity and a harder boundary-specific form of specificity that the preceding four-direction pilot lacked.

Selection used a fixed PCG64 stream of 250,000 ten-clause proposals from the complete native 40-clause grammar and a response-blind, domain-separated hash order. It excluded all 9,469 token rows in the preceding native-query ledgers. The selected base array has $16 \times 5 \times 32 = 2,560$ assignment-stratum rows representing 2,543 distinct formulas. For each of 118 labelled assignment–transposition edges and each of the five strata, the transformed arm retains the first eight hash-ordered eligible base cases whose transformed token differs and is historically novel. It therefore takes one quarter of each base cell and supplies 4,720 linked transformed rows. Deduplication leaves 7,252 native checkpoint inputs. No learner is fitted. An independent verifier checks 2.5 million clause identifiers, eight million truth-count cells, every base and transformed semantic label, all 7,252 orbit identifiers, all near/far cells and the complete historical exclusion before inference. A separate eight-row reload reproduces every preactivation, threshold bit and task argmax.

The primary threshold-free score is AUROC for each unit's 64 positive and 96 negative base cases. The frozen protocol grouped three ranking gates and one source-threshold specificity guard under a combined "broad-discrimination" decision. A separate source-threshold criterion requires at least 24/32 detections in each positive stratum for every unit, at most 3/32 false positives in every negative stratum for every unit, and pooled threshold balanced accuracy at least 0.85 for at least fifteen units. The specificity guard is thus shared by the two conjunctions. These are study-defined decision rules, not guarantees made by Palumbo et al.; the original grouping is retained for provenance but is not evidence for two independent failures.

Pre-specified component	Observed result	Decision
Sixteen-assignment macro AUROC ≥ 0.95	0.991394	Pass
Every unit AUROC ≥ 0.80	Range 0.961100–1.000000	Pass
At least 15/17 unit AUROCs ≥ 0.90	17/17	Pass
Shared threshold-specificity guard: no multi-solution-negative false-positive rate > 0.10	Units 10 and 81 each 4/32	**Fail**
Fidelity positive cells: every cell at least 24/32	Failures for units 81, 261, 411 and 416	**Fail**
Fidelity pooled balanced accuracy ≥ 0.85 for at least 15/17 units	16/17	Pass

Under the original conjunction, the protected combined criterion is therefore **not met** only because the shared threshold-specificity guard fails; every threshold-free ranking gate passes. The source-threshold fidelity criterion is also **not met** because of that shared guard and the positive-cell failures. The formula-orbit bootstrap interval for assignment-macro AUROC is [0.988258, 0.994147]. It resamples full variable-permutation orbits within solution-count classes and is descriptive for this constructed mixture; it is not a checkpoint- or neuron-population interval.

The conjunction was stringent. Under an illustrative independent homogeneous-cell model, if every positive cell had true sensitivity 0.80, all 34 positive cells would pass with probability 0.0015; at sensitivity 0.90 the probability is 0.894. Using the observed multi-solution near-negative rate 23/272, a 16-case near half would contain at least four false positives with probability 0.041, and at least one of 17 such cells would do so with probability 0.507. These calculations deliberately use assumptions that are false for this dependent engineered bank; they show why the gate outcome should not carry the main inferential weight. Under the same illustrative independent-Bernoulli sampling model, exact 95% binomial intervals would be [0.035, 0.290] for 4/32 and [0.479, 0.729] for unit 81's pooled 39/64 positive detections. These are sampling-model sensitivity calculations, not justified population confidence intervals for the engineered dependent bank. Unit 81 has the clearest observed sensitivity deficit. The 23/32 cells for units 261, 411 and 416 are each one observation below the study gate and should be read as boundary failures, not sharp discontinuities in evidence.

4.1.2 Unit-level results and localisation of errors

Unit	Published assignment	AUROC	Sole-positive TP	Multi-positive TP	FP: UNSAT / different-sole / excluding-multi	Pooled balanced accuracy
10	TFFFF	0.981	32/32	31/32	0/3/4	0.956
41	FFFFF	0.996	32/32	30/32	0/1/0	0.979
55	TTTFF	0.979	24/32	25/32	0/1/1	0.872
77	FFTTT	0.992	32/32	32/32	0/3/2	0.974
81	FTTTF	0.961	19/32	20/32	0/0/4	0.784
185	FFFTT	1.000	32/32	32/32	0/0/0	1.000
224	FFFFT	1.000	31/32	32/32	0/0/0	0.992
245	FTFFT	1.000	31/32	32/32	0/0/1	0.987
261	TTFTT	0.993	27/32	23/32	0/0/0	0.891
291	TFTTT	1.000	32/32	31/32	0/0/1	0.987
317	TFFFT	1.000	32/32	32/32	0/0/0	1.000
326	TTTTF	1.000	32/32	32/32	0/0/1	0.995
380	TFFTT	1.000	32/32	31/32	0/0/0	0.992
411	TTFFT	0.974	23/32	28/32	0/2/3	0.872
416	TTTFF	0.981	23/32	26/32	0/2/2	0.862
492	TFTFT	0.990	31/32	30/32	0/1/3	0.956
495	FFFTF	0.995	32/32	29/32	0/0/1	0.971

The threshold detects 497/544 sole-solution positives and 496/544 multi-solution positives. False positives are 0/544 on UNSAT formulas and 0/544 on far satisfiable negatives, but 36/544 on Hamming-one negatives. Thus every threshold false positive in these balanced base cells is local to a formula that has a nearby solution; this sentence does not cover the 1,000 full-bank false positives recorded by the readout audit. Per-unit near-negative counts are 7, 1, 2, 5, 4, 0, 0, 1, 0, 1, 0, 1, 0, 5, 4, 4 and 1 out of 32. A common-rate Pearson dispersion statistic is 40.34 on a nominal 16 degrees of freedom, but no reference p-value is used: unit is confounded with assignment-specific formula composition, rows share orbits, and units 55 and 416 reuse the same formulas. The two duplicate TTTFF units are separate response measurements on that shared bank, so pooled counts are descriptive rather than independent trials.

The same locality appears without a threshold. Comparing every unit's 64 positives with its negatives yields 970 strict ranking inversions: 935 (96.4%) involve near negatives, 10 involve far negatives and 25 involve UNSAT negatives. There are no tied pairs. Against the 32 near negatives alone, mean unit AUROC is 0.9731 and ranges from 0.8896 for unit 81 to 1.000 for five units. At a threshold above every observed near negative, retained sensitivity ranges from 9/64 for unit 81 to 64/64 for five units; allowing one observed near false positive raises some units but leaves unit 81 at 10/64. These estimates distinguish a misplaced source threshold from genuine finite-bank overlap.

A separate post hoc competition against all 32 possible assignment labels sharpens the positive result: the published assignment is the best-separating candidate for every one of the 17 units on the complete 7,252-input bank. Using preactivation positive-direction AUROC, its margin over the runner-up ranges from 0.027 for unit 81 to 0.215 for unit 224, and every runner-up assignment is one Hamming bit away. The competition AUROCs use different positive populations from the protected base AUROCs, and the bank was engineered around the 16 published labels rather than the 16 unlabelled assignments; the 17/17 result is therefore label-relative evidence on an asymmetric bank, not an unbiased prevalence test. Using post-ReLU two-sided AUROC on the complete bank, thirteen nominated units rank first among all 512 for their label and four rank second: unit 81 is behind 334 for FTTTF, 261 behind 374 for TTFTT, 411 behind 201 for TTFFT, and 416 behind 55

for the shared TTTFF label. Under the protected preactivation positive-direction score, the 13/4 split is unchanged but unit 416 ranks first and unit 55 second for TTTFF; the scoring basis therefore accompanies every rank claim. Thus units 374 and 201, used as activity-profile comparisons in the earlier pilot, are stronger same-assignment axes rather than semantic nulls. The 32-by-34 matrix in Figure 3 reports the complete source-bank competition.

For a fixed assignment in this grammar, the assignment satisfies a formula exactly when none of ten assignment-specific clause types occurs. Palumbo et al.'s weighted-count account therefore supplies a direct antecedent: the number of clauses violated by the target assignment is a more specific prediction than a generic literal Boolean-detector account. A post hoc regression of native preactivation on 1,088 satisfiable-negative unit-formula rows keeps every variable-permutation orbit in one of five folds. Unit means plus categorical violation count attain cross-validated $R^2 = 0.655$, compared with 0.348 for unit means plus categorical nearest-solution distance; combining both reaches 0.688. Thus adding count after distance contributes 0.340, while adding distance after count contributes 0.033 on this bank. A unit-specific linear-count model reaches $R^2 = 0.721$; this is an R^2, not a slope. The training fold labelled 4 is rank deficient by one column in the count and combined designs; the global category columns are retained, so a category absent from a training fold receives coefficient zero and held-out predictions use the remaining terms. Because the bank explicitly balances near/far distance while allowing count to vary, these values describe available variation rather than adjudicating mechanisms.

Within violated-clause count one, substantial structure remains. Two-neighbour near negatives fire on 9/22 events and have mean score 0.221; one-neighbour near negatives fire on 25/237 and average -0.213; matched UNSAT and far rows fire on 0/122 and 0/44 and average -0.399 and -0.768. Their strict positive-pair inversion counts are respectively 184, 657, 22 and 5. In this two-literal grammar, two distinct Hamming-one satisfying neighbours logically force every target-violated clause to be the unique clause type on the two flipped variables. One distinct violated type is therefore a necessity, not an empirical finding. The within-count ordering shows that count alone is incomplete and motivates a crossed dissociation bank varying count, distance, neighbour multiplicity and flip damage. False positives are also uneven across the exclusively nearest flipped variable, with counts 0/100, 11/101, 8/99, 0/108 and 8/112 for x_0 through x_4. These are post hoc mapping diagnostics, not pre-specified variable or multiplicity effects.

Task accuracy and unit-label fidelity are different objects. The checkpoint predicts 7,215/7,252 formulas correctly: errors are 2/1,453 UNSAT, 35/2,900 sole-solution and 0/2,899 multi-solution cases. Across the 2,720 dependent base unit-formula events, 2,580 have both labels correct, 130 have a threshold-label error with a correct task prediction, nine have a task error with a correct unit label, and one has both errors. Stratification explains why this cross-tab is unsurprising: only 1/47 sole-solution misses coincides with a task error, while 0/48 multi-solution misses and 0/36 satisfiable-negative false positives do. Multi-solution misses can be compensated by other solution units, and every false positive occurs where SAT is already the correct task output. The much lower task-error base rate and repeated use of each formula still make these counts descriptive; they do not establish that threshold errors are causally harmless. A later task-error localisation finds 9/147 errors (6.12%) on the three weakest single-axis assignments, versus 26/2,753 (0.94%) on the other sole-solution assignments. The linked rows are not independent binomial trials, so no sampling interval is claimed. Assignment composition is confounded, and FTTFF has 4/45 errors despite a strong dedicated axis, so this is an exploratory association rather than a causal error explanation.

4.1.3 Variable transpositions are a dependent consistency check

On transformed inputs, mean pooled unit AUROC is 0.994898 (median 0.998372; minimum 0.963469). At threshold, fixed-assignment edges detect 1,269/1,376 positives with 22/2,064 negative firings; assignment-switching edges detect 561/624 positives with 10/936 negative firings. Every transformed false positive again lies in the near stratum: 22/374 for fixed edges and 10/184 for switching edges,

versus zero in the 1,002 and 440 far cases and the 688 and 312 UNSAT cases. A variable transposition is a Hamming isometry and does not change assignment-space difficulty. The retained composition differs after transformed-arm eligibility, historical-novelty filtering and the fixed eight-case quota. Their separate contributions to the imbalance have not been diagnosed: near cases form only 18.1% of all fixed negatives and 19.7% of all switching negatives, versus 33.3% of all base negatives. Among satisfiable negatives alone the corresponding shares are 27.2%, 29.5% and 50.0%. This observed selection imbalance means pooled transformed AUROC should not be compared directly with the base result. Like-for-like mean near AUROCs are 0.979 fixed, 0.973 switching and 0.973 base; fixed and switching cell minima are 0.862 and 0.866. Because the formula generator is permutation-symmetric, the transformed rows are linked to base rows, and units 55 and 416 duplicate one semantic label, these figures are a consistency check on semantic transport. They do not establish architectural equivariance, equal raw activations under renaming, independent replication, or a shared implementation across units.

The most defensible reading is therefore graded. In this checkpoint and selected cohort, each published singleton assignment strongly separates its nominated unit's scores from balanced negatives: all seventeen protected AUROCs exceed 0.96, the hard-boundary mean is 0.973, and units 185 and 317 are error-free at the source threshold on the balanced base cells. That statement coexists with finite-bank overlap for several units, material unit-to-unit threshold-fidelity differences and the shared threshold-specificity failure. Describing these units simply as Boolean detectors, as uniformly "validated", or as computing satisfiability would exceed the evidence.

4.1.4 A new response-blind source-grammar bank

The protected bank deliberately stresses semantic strata, so its distribution is not the source sampling distribution. We therefore generated a separate, response-blind bank of 4,096 formulas, balanced at 2,048 SAT and 2,048 UNSAT. Each formula has ten distinct non-tautological clauses, each involving two distinct variable identities, from the source's forty-clause grammar. Before inference we excluded 365,753 historical clause multisets from 51 declared arrays. Balancing, deduplication and history exclusion mean this is not an independent sample at natural prevalence. Unknown training inputs could not be excluded. Labels, checkpoint and threshold 0.5 remained fixed; nothing was fitted or rematched.

New-bank quantity	Result
Native full-vocabulary argmax accuracy	4,087/4,096 (99.7803%)
SAT and UNSAT accuracy separately	2,042/2,048 and 2,045/2,048
Assignment-macro AUROC; minimum unit AUROC	0.997477; 0.989887
Unit 55 threshold detections; false positives	83/113; 34/3,983
Unit 81 threshold detections; false positives	88/125; 50/3,971
Unit 416 threshold detections; false positives	90/113; 60/3,983

Strong semantic ranking thus transfers to this declared distinct-clause distribution, while appreciable threshold misses remain. The original 32-case-cell criteria were not reapplied to different denominators. This is new-input evidence for the same checkpoint on the same host, not an independent-checkpoint replication. Supplement Appendix K gives all seventeen units, input-only history repair, frozen provenance and independent recounts.

4.2 Complete-layer assignment alignment and effective width

Palumbo et al.'s report J contains 34 final-MLP units selected because each unit's largest direct effect on the SAT output exceeded 0.1 on the source training data. The recorded parameters place

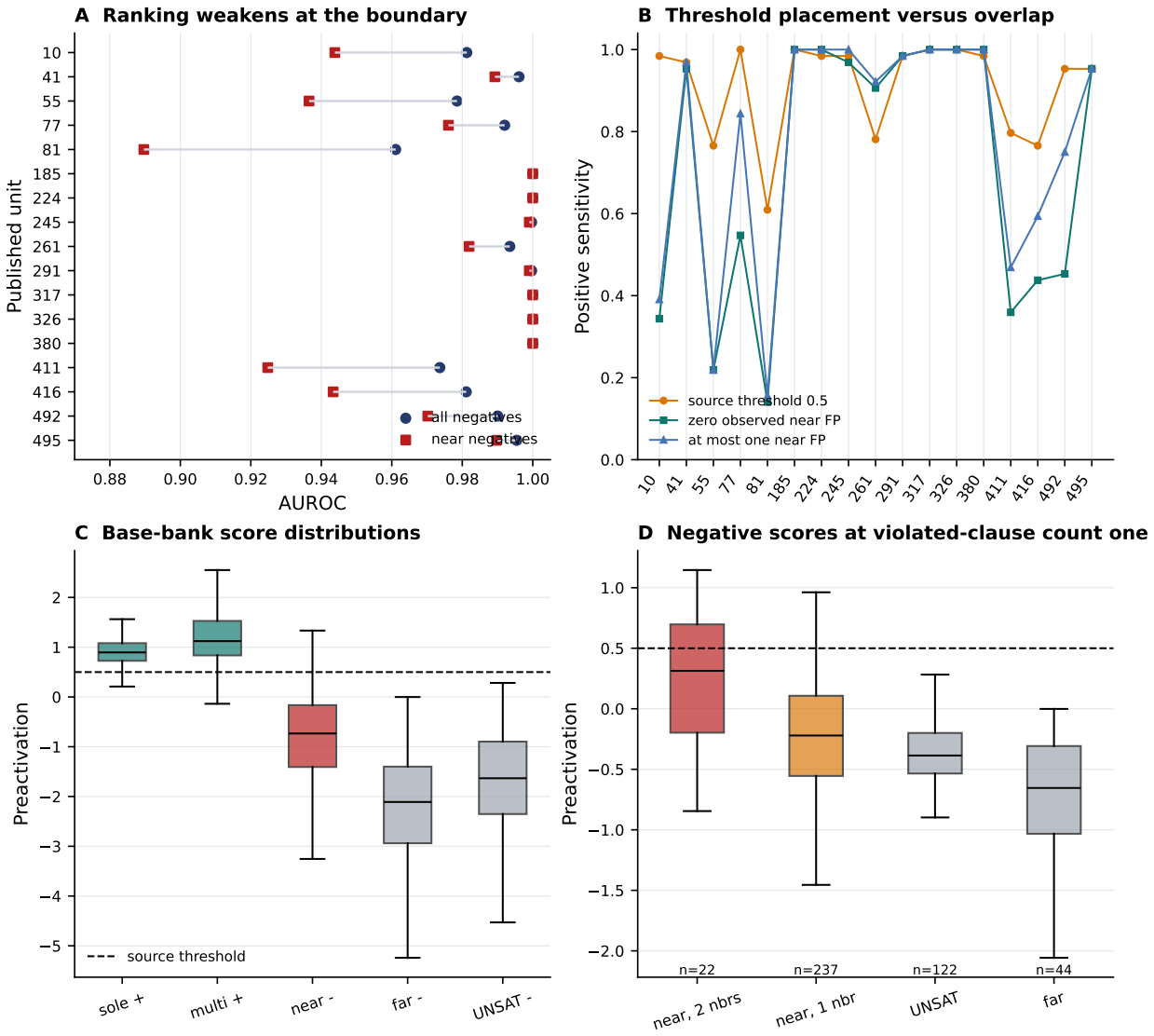


Figure 1: Protected and post hoc views of the all-unit SAT label audit. **A:** Protected AUROC against all negative strata and post hoc AUROC against Hamming-one satisfiable negatives only. **B:** Positive sensitivity at the source threshold and at the most permissive observed thresholds allowing zero or one near-negative false positive. **C:** Pooled base-bank score distributions; all source-threshold false positives occur in the near stratum. **D:** Negative scores at fixed violated-clause count one, showing residual separation by solution geometry. Panels B–D are finite-bank diagnostics selected after the protected result. The denominators share formulas and one checkpoint and do not estimate a unit population or establish a causal mechanism.

477 of the 478 units outside J at scale at most 3.74×10^{-12}; the remaining unit is silent on this bank. Localisation to J is therefore largely a mechanical consequence of the source's effect threshold and the trained weight pattern, not evidence that a selector isolated 34 units from a comparably energetic remainder. The source then fitted shallow logical rules to those units. This selection resembles direct-logit analysis of transformer feed-forward outputs, where unit contributions form additive vocabulary-space updates (Geva et al., 2022). The seventeen-unit cohort used above has a designated singleton assignment for each unit, while the source's broader decision-tree inventories can associate some of those same units with one to three Hamming-adjacent assignments. The other seventeen units have conjunction, exclusion or multi-assignment rules. The label audit fixes the designated singleton labels; the full-layer study asks how the complete state is organised.

4.2.1 Protected full-layer comparison

The protected extraction covers the same 7,252 inputs, with the earlier seventeen columns and task logits treated as anchors. Every anchor is bitwise identical. The historical eight-row fresh-load replay reproduced its anchors. The corrected release additionally re-executes all 7,252 inputs from a clean extraction, reproducing preactivations, post-ReLU states, both retained residual streams and logits bitwise on this host. On the complete bank, 442/512 units are exactly zero, 70 are numerically nonzero, and the post-ReLU matrix has numerical rank 34.

Before full-layer model access, the study fixed 63 Haar-distributed 512-dimensional rotations and two primary one-to-one matching statistics. The first matches all 32 assignments; the second matches the sixteen assignments absent from the published singleton-label set. A secondary frozen licensing condition restricts positives for each of those sixteen assignments to SAT formulas with no covered solution. The raw finite-reference ranks are 1/64 for both primaries; the frozen two-test procedure transforms those ranks to 1/32. This is a pre-specified decision rule on finite-ensemble extremeness ranks, not a Holm familywise-error guarantee or a randomisation p value, because sparse nonnegative native coordinates and dense signed rotations are not exchangeable. With 63 references the rank is saturated; native-minus-best margins carry the effect-size information.

For clarity, let L be the selected assignment labels, B the specified row population, and x_j a candidate coordinate. Put $A_{\ell j} = \mathrm{AUROC}_B(x_j, \mathbf{1}[\ell \models \varphi])$. The reported matching mean is

$$M(L,B) = \max_{\pi : L \hookrightarrow \{1,\dots,d\}} \frac{1}{|L|} \sum_{\ell \in L} \max\{A_{\ell,\pi(\ell)}, 1 - A_{\ell,\pi(\ell)}\}.$$

Thus labels receive distinct coordinates, while score orientation may be chosen separately for each label-coordinate pair. AUROC is the positive-negative ranking probability with half credit for tied scores; the implementation uses exact-equality average ranks, so a constant coordinate scores 0.5. Maximum-weight matching uses a deterministic Hungarian implementation; tied optima need not identify a unique scientific axis. The protected comparisons use all 512 post-ReLU coordinates, or all 512 coordinates after right multiplication by the fixed orthogonal rotation, without restricting matches to source units. The all-32 and uncovered-16 assignments are optimised separately on all 7,252 rows, including UNSAT negatives. The exclusive secondary restricts the entire population to SAT formulas with no covered solution: a given uncovered assignment's positives satisfy that assignment and its negatives are the other rows in this restricted population. Retrospective comparisons below instead use their explicitly named 34- or 70-coordinate state and transformed scores; they apply the same matching functional. The earlier all-unit preactivation AUROCs use the source-positive direction and are not this two-sided matching statistic.

Protected matching target	Native mean AUROC	Full-space Haar mean	Best of 63	Native minus best	Reference rank
All 32 assignments	0.984233	0.953018	0.957967	+0.026266	1/64
Sixteen labels outside the singleton set	0.974041	0.949115	0.956669	+0.017372	1/64
Exclusive-positive secondary	0.978782	0.973042	0.977960	+0.000823	1/64, licensing secondary

The protected control labels reverse direction: native versus Haar ranges are 0.543 versus 0.644–0.652 for clause-presence matching, 0.564 versus 0.785–0.929 for the SAT label, 0.534 versus 0.590–0.633 for sole-solution status, and 0.568 versus 0.762–0.901 for multi-solution status. SAT's reversal is expected from the dense positive readout direction across all live units.

The count descriptives in this protected comparison are partly mechanical. Native coordinates supply only 34 effective axes, whereas a 512-dimensional rotation produces hundreds of dense mixtures of the same numerical-rank-34 signal. Native therefore has fewer axes above 0.90 and 0.95 than every full-space rotation, while having many more above 0.98 and 0.99. The weakest native assignment axis also loses to every full-space rotation. The three weakest assignments—FTFFF, TFTFF and TTTFT—are exactly the three omitted by the source's two-leaf dedicated-rule cover. The exclusive-positive condition passed by only 0.000823, about 0.32 of the rotation standard deviation, so the 1/64 rank must not be read as a large effect. Its threshold was binding even though the frozen protocol described its inferential status as descriptive.

The protocol's authoritative conclusion is reproduced verbatim:

> Native coordinates exceed the fixed Haar references on both primary matching statistics; the 32-assignment primary includes 16 assignments selected for published native units. The native-over-Haar matching advantage is also observed in the descriptive exclusive-positive secondary (uncovered assignments, SAT formulas with no covered solution; matched axes may include published units; not multiplicity-controlled). This is a within-checkpoint, within-bank single-axis-separability comparison against Haar references whose exchangeability null is generically false for nonnegative sparse ReLU coordinates; it does not separate assignment-specific alignment from generic coordinate privilege (see the clause-presence and SAT/solution-count control gaps) and is not evidence that any coordinate is causally used.

4.2.2 Retrospective effective-state and geometry controls

Opening the full result reveals that J is not a small sub-bank inside a broadly active layer. Its 34 columns contain a squared-norm share of one to double precision (the stored value 0.9999999999999998 reflects roundoff); the other 36 numerically nonzero columns contribute a share of 6.98×10^{-53}. Their largest activation is 1.03×10^{-24}. The singular spectrum has 34 values from 51.45 down to 4.08, followed by values at or below 5.97×10^{-15}; the rank remains 34 for relative tolerances from 10^{-12} through 10^{-4}. Thus J is the effective live state of this final MLP on the bank. A zero-forward parameter audit adds an important qualification. Of the 478 non-J units, 477 have a maximum input-norm, bias or output-norm scale at or below 3.74×10^{-12}; unit 386 retains modest weights (input norm 0.0544, output norm 0.0848) but never activates on this bank. The effective-state

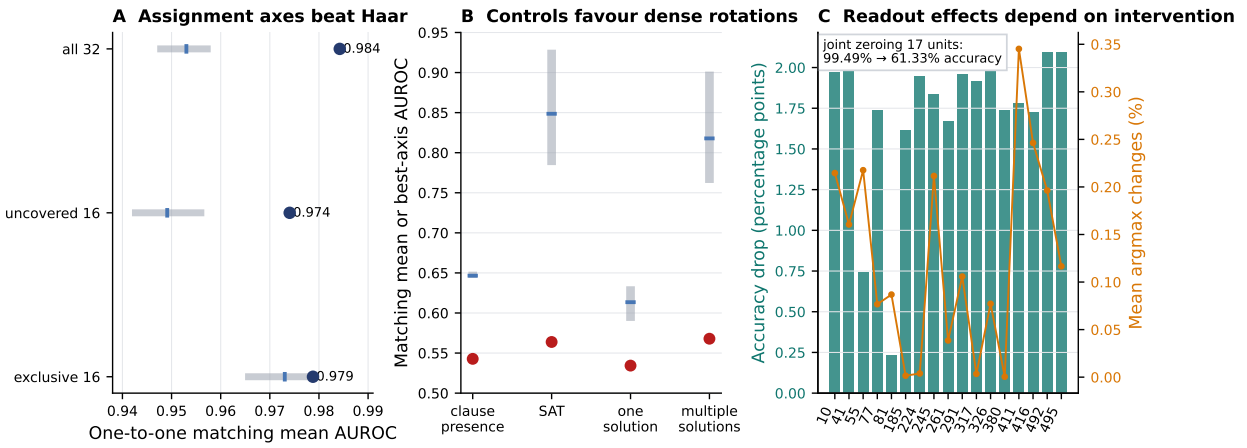


Figure 2: Protected full-layer native-basis and exact-readout results. **A:** labelled dark points are native one-to-one assignment-matching means; grey segments show the full range over 63 fixed full-space Haar rotations and blue ticks their means. The exclusive-positive result is a frozen licensing secondary, outside the two-primary multiplicity procedure. **B:** four differently structured controls reverse direction. **C:** teal bars show the task-accuracy drop from individually zeroing each published unit at the exact final affine cut; the orange line shows mean argmax changes under within-stratum activation resampling. Joint zeroing of all seventeen units lowers accuracy from 99.49% to 61.33%. The full-space basis comparison is the protected historical test; the geometry-matched retrospective controls below determine its interpretation.

identity therefore accompanies near-zero parameters outside J, with one bank-silent exception; the training cause of that pattern is not established, and it is not an input-independent theorem about every possible formula. Post hoc exclusion of J collapses assignment matching because it removes essentially the entire observed state, not because alignment was localised against a substantial non-J alternative.

This structural correction does not erase the native-axis result. The native all-32 mean is 0.984233 in every row below. Repeating the one-to-one analysis *inside the live state* gives the following retrospective descriptive comparisons:

Reference family, 63 draws	Reference mean	Best reference	Native minus best	Weakest native minus best
34-dimensional Haar rotations within span(J)	0.901504	0.915146	+0.069087	+0.021557
70-dimensional Haar rotations of numerically active columns	0.924260	0.932969	+0.051265	−0.011003
Standardised 34-dimensional Haar rotations	0.902659	0.914323	+0.069910	+0.031780
Dense nonnegative mixtures of J	0.904640	0.915189	+0.069045	+0.021916
Two-sparse nonnegative mixtures of J	0.928814	0.954067	+0.030166	+0.098758

The native mean exceeds every reference in all five families. Within the selected 34-column live span, the weakest native matched axis exceeds the weakest matched axis of every raw rotation, every standardised rotation and every dense positive mixture for the all-32 match. For the uncovered-assignment subset, one standardised rotation has minimum 0.8843 versus the native 0.8809; the mean advantage is not a universal per-axis advantage. The same comparison on the 2,543 unique base formulas yields native 0.985665 versus a best 34-dimensional rotation of 0.919789. The 70-coordinate reference supplies 70 directions in an effectively rank-34 state, an intermediate degree of overcompleteness. Dense positive mixtures use independent exponential entries; two-sparse mixtures choose two distinct input coordinates without replacement per output column and give them independent exponential weights. Each column is normalised to unit Euclidean length. These are mixtures, without a required invertibility condition, and remain constructed references rather than exchangeability nulls or learned alternatives.

Two controls constrain the interpretation. Independently permuting each assignment-label column preserves its exact 3.5–7.3% prevalence but destroys its semantics; native mean matching is then 0.5159 and loses to every in-span rotation (range 0.5321–0.5369). For the 32 labels $q_a = 1$, meaning that assignment a violates exactly one clause, native mean matching is 0.5530 and again loses to every rotation (0.6032–0.6151). Clause-violation count is semantically related to the source rules, so this is a monotonic-band sensitivity rather than a semantic null. Native coordinates therefore do not win on the shuffled-label control or this broader one-violation band. The shuffled-label reversal is partly mechanical because sparse post-ReLU axes create many average-rank ties and consequently less extreme noise AUROCs than dense continuous rotations; the structured clause and solution-count controls carry more semantic weight.

A supervised learned comparator supplies an upper descriptive benchmark. A five-fold ridge least-squares probe ($\lambda = 1$, intercept unpenalised), grouping variable-permutation orbits within folds and standardising from training folds only, on the 34-dimensional state reaches mean AUROC 0.9956 and minimum 0.9842, outperforming the native single-axis match. For 29 of 32 assignments, one native unit is within 0.018 of that probe. For the three assignments with weak native single axes, probe AUROCs are 0.9898 (FTFFF), 0.9924 (TFTFF) and 0.9859 (TTTFT), compared with best-unit AUROCs 0.9029, 0.8809 and 0.8941. The orbit-grouped cross-validated linear probes predict those assignment labels more accurately than the best individual native axes, without establishing exact linear representations. This distinguishes a clean native report from a claim that every assignment has its own Boolean neuron.

A retrospective 32-label competition across all 34 source units finds that every unit's best assignment AUROC is 0.946–1.000, including all seventeen units whose published source rule is not singleton; collectively their best matches cover 29 assignments. Ranking is not purity: the fraction of each unit's summed activation mass on its best label ranges from 0.595 to 0.994 (median 0.893), with eighteen units below 0.90. The native axes are therefore graded local-region SAT-evidence directions rather than uniformly atomic assignment detectors. The engineered bank oversamples hard negatives around the sixteen source-covered labels, so purity and covered/uncovered comparisons are not estimates for the source distribution. All 32 axes selected by the protected match belong to J, and each matched assignment appears in that unit's source tree inventory. Two matches in the supposedly uncovered half use published singleton units 416 and 411; the protected second primary excluded their designated labels, not their axes or the broader public source rules. Excluding the singleton seventeen leaves fourteen of sixteen uncovered matches unchanged; only TFTFF and TTTFT collapse, from 0.881 to 0.542 and from 0.894 to 0.641. These source-linkage facts are retrospective consistency checks, not held-out discovery.

Learned decompositions further limit the native-basis claim. Across five seeds, 34-component NMF attains mean one-to-one assignment matching 0.9862–0.9865, slightly above native 0.9842; its weakest match ranges from 0.8807 to 0.8879, versus native 0.8809. It recovers 32 of 34 synthesis axes at absolute cosine at least 0.99 (mean matched cosine 0.9626), but misses axes 261 and 374, with minimum matched cosine zero. Its relative Frobenius reconstruction error is 0.03145. Because the

input state X is nonnegative, $X = XI_{34}$ is an exact feasible NMF: the fitted solution is not globally optimal for its reconstruction objective. A retained-array diagnostic finds 1,325 rows with exactly one active native unit, supplying 2–124 such examples for each of all 34 axes. This makes classical separability a pertinent alternative explanation of axis recovery (Donoho and Stodden, 2003; Arora et al., 2012), without claiming that the local solver recovered an identifiable latent generative process. NMF of neural activations also has established interpretability antecedents (Olah et al., 2018).

The observed pure-coordinate rows give a more specific exact statement. Suppose $X = WH$ with both factors nonnegative and exactly 34 components. A positive row of X supported only on coordinate j must use at least one nonzero row of H supported only on j: nonnegative contributions cannot cancel on any other coordinate. The 34 observed coordinate rays therefore require 34 distinct components, exhausting the component budget. Every exact such factorisation recovers the native rays up to positive scaling and permutation. This elementary deduction is about the retained matrix and exact factorisation; it is not a claim about unseen latent variables. The fitted solution has nonzero reconstruction error, so the deduction does not require it to recover every axis exactly or to give the native score ranks.

Per-assignment attribution locates the aggregate improvement in scores of components whose synthesis directions remain close to native axes. The two poorly recovered components contribute a small negative mean difference, -0.000025 to -0.000028. For example, in seed zero the FTFFF match rises from 0.9029 to 0.9463 even though its synthesis direction has cosine above 0.999999 with native unit 48. Similar synthesis directions do not imply identical component scores or identical rank ties. The complete per-assignment changes accompany the diagnostic; the fitted factorisation's matching score and its reconstruction objective must be kept separate.

A 32-component NMF loses one assignment badly (minimum 0.5416); 34-component independent component analysis (ICA) scores below native. All decompositions are fitted and scored on the same retained state using seeds 0–4, so this is not held-out generalisation. NMF uses NNDSVD initialisation with small random fills, Frobenius loss, a maximum of 2,000 iterations and tolerance 10^{-6}; the 34-component runs stop at 193–194 iterations without convergence warnings. These seeds test local numerical stability, not broadly distinct initialisations. ICA uses unit-variance whitening and the same maximum and tolerance. Its synthesis directions are the transpose of the mixing matrix, and reconstruction adds the fitted mean; the corrected relative error is $1.2\text{–}1.4 \times 10^{-15}$. The earlier unmixing-matrix reconstruction was wrong; the component-score matching was unaffected. A fixed mini-batch dictionary attempt (34 atoms, batch size 512, L1 penalty one, maximum 1,000 iterations) stopped after only 2–6 iterations with relative reconstruction error 0.693–0.698. It is retained as an unsuccessful comparator, excluded from any representation ranking. The learned methods have no protected common complexity budget. Their evidence supports assignment alignment and partial recovery of native axes, not a uniquely optimal or generally identifiable native basis.

4.2.3 Exact readout accounting: additive logits and redundant margins

For the last MLP state h, the logits are exactly $z = rW_U + b_{out}W_U + hC$. There is no interaction among units at this cut: zeroing a set subtracts the sum of its unit-level logit contributions. All 34 effective units have positive SAT-minus-UNSAT coefficients, ranging from 5.813 to 11.963. The final MLP therefore supplies positive SAT evidence; the residual-plus-bias path predicts UNSAT on almost every bank row when all 34 contributions are removed.

Individually zeroing the published seventeen changes 17–153 argmaxes per unit. Jointly zeroing them changes 2,771/7,252 decisions and lowers accuracy from 99.49% to 61.33%. Their individual change counts sum to 2,131 and their union covers 2,071 rows. The protected baseline, individual-zero and grouped-zero checks record no top-two logit gap below 10^{-4}, so their argmax changes do not depend on near-ties at that guard. The joint count is larger because several additive positive contributions can jointly cross an argmax margin when no single removal does. This is redundancy after thresholding logits to a decision, not interaction in the readout. The pair 81/334 illustrates

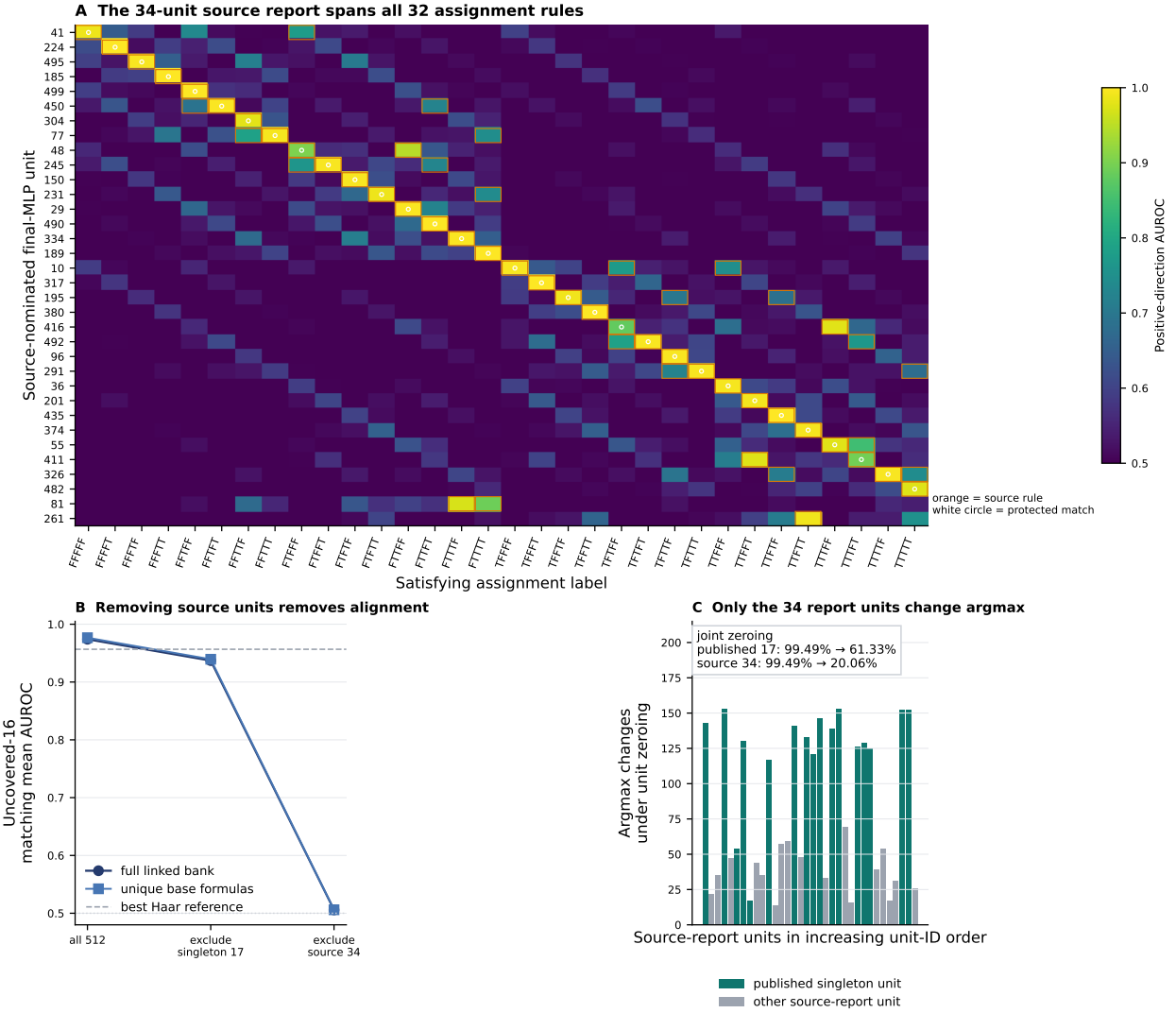


Figure 3: Semantic, geometric and readout views of the effective final-MLP state. **A:** post-ReLU positive-direction AUROC for the 34 source-report units against all 32 satisfying assignments. Orange boxes mark assignments in the source tree inventories; white circles mark protected one-to-one matches. **B:** retrospective uncovered-assignment matching on the complete linked bank and unique base formulas. Removing J removes effectively all post-ReLU energy, so the resulting chance-level score is structural. **C:** exhaustive individual zeroing on the 7,252-input bank. The same 34 live units are the only units that change a task argmax. Joint zeroing values are exact at the final affine cut.

it cleanly: their individual counts are 17 and 69, their union is 78, and joint zeroing changes 140 decisions. The earlier 55/416 example is subadditive relative to the individual-count sum and is no longer used to support this point.

Jointly zeroing all 34 changes 5,764 decisions and reduces accuracy to 20.06%, approximately the bank's 20.04% UNSAT prevalence. The residual path predicts UNSAT on 7,250/7,252 rows and produces no non-answer argmax. The 34-unit source state is exactly zero on 992/1,453 UNSAT rows and on no SAT row, so the residual result mainly recovers the class prior-like UNSAT default rather than a second solver. Among baseline-correct cases it flips 2,720/2,720 covered-only cases, 2,050/2,050 cases with covered and uncovered solutions, and 992/994 uncovered-only cases from correct to incorrect; the two reverse changes are UNSAT inputs. Because J is the whole effective final-MLP state on the bank, this ablation identifies the dominant direct contribution at the final-MLP output cut, not an upstream source or route. The readout is assignment-agnostic: it sums positive evidence and does not itself use assignment identity.

The graded claim rests on contribution magnitudes rather than joint-count arithmetic. Across the seventeen published units, mean SAT-minus-UNSAT contributions on source-threshold false negatives range from 1.72 to 4.76 logits. Where false positives exist, their means range from 5.14 to 8.75 logits. Some of these rule-inconsistent and sub-threshold activations support correct SAT decisions; zeroing them can turn a correct prediction into an error. The thresholded logical report is therefore a useful coarse description, while the affine readout uses continuous positive evidence. A five-fold orbit-grouped fixed-readout substitution makes the calibration loss operational: replacing each live activation by its threshold bit times a training-fold median preserves 98.39% of native argmaxes but lowers task accuracy from 99.49% to 98.15%; sole-solution accuracy falls from 98.79% to 95.52%, while UNSAT accuracy rises to 100%. This measures the loss for the specified unrefitted substitution; it does not establish the minimum loss over bit-based decoders. A separate exploratory predictive comparison uses the same five outer orbit-hash folds for both representations. Each training fit standardises its inputs, and an inner leave-one-of-four-fold-out procedure chooses the L2 parameter C from $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$ by total validation errors, breaking ties toward smaller C. All scaling and parameter selection exclude the outer assessment fold. The refitted classifier makes 64 errors from the 34 bits (99.12% accuracy) and 26 from continuous activations (99.64%). There are 49 cases correct only for the continuous classifier, 11 correct only for the bit classifier, and 15 both miss. Repeating the complete nested procedure without standardising either representation gives 64 and 21 errors (99.12% and 99.71%). The direction of this predictive difference survives both matched preprocessing choices; its magnitude depends on fitting. The original 99.68% continuous result used a fixed $C = 1$ and unmatched preprocessing and is superseded as the headline comparator. These classifiers predict task truth from the state alone; the native-readout substitution also retains the residual stream. Their accuracies therefore do not decompose a single calibration loss. The continuous truth classifier's 26 errors are fewer than the native checkpoint's 37, which illustrates that it predicts task truth rather than reproducing the checkpoint; its separately reconstructed agreement with native argmaxes is reported below. Neither supervised comparison is a causal abstraction, an optimum over all decoders or an information-loss bound.

Within-assignment SAT/solution-count resampling changes only 0.0004%–0.3452% of argmaxes on average across the published units. This preserves task labels and much of the assignment signal, so it is a sanity check rather than a mechanistic control. At this final affine cut the total intervention is exactly the direct-logit effect and no downstream self-repair is possible; the compensatory effects documented for earlier-layer ablations therefore do not apply at this cut (McGrath et al., 2023). Zero occurs naturally on 992 UNSAT states. Applying it to another row while holding that row's residual fixed can nevertheless create an off-manifold joint state; it is not a generally canonical ablation value (Li and Janson, 2024). None of these results identifies how the activations arose or whether the same route is used upstream.

The fitted truth classifiers can also be compared with the native decision, without confusing the two targets. On the original 7,252-formula bank, the standardised bit and continuous classifiers agree

with native full-vocabulary argmax on 7,205 and 7,221 rows (99.3519% and 99.5725%), respectively. The unscaled sensitivity gives 7,205 and 7,224 agreements. These predictions were reconstructed retrospectively from the freshly regenerated complete binary error sets, not obtained by fitting a native-target classifier. This does not recover logits or deploy a fitted decoder on the new bank; Appendix K specifies the reconstruction.

4.3 The edit target is not a function of current truth or the bit report

The following ordered formulas share the sole satisfying valuation 10000, with bits ordered $x_0, \ldots, x_4$. In the table, ij denotes $x_i \vee x_j$ and a bar negates that literal.

Slot	φ_A	φ_B
0	$0\bar{1}$	$0\bar{1}$
1	$\bar{0}\bar{4}$	$\bar{2}3$
2	04	$\bar{0}\bar{1}$
3	$3\bar{4}$	$\bar{1}\bar{3}$
4	$\bar{0}\bar{1}$	$\bar{2}\bar{3}$
5	$\bar{1}2$	$\bar{3}4$
6	$\bar{0}\bar{2}$	$0\bar{3}$
7	$1\bar{3}$	03
8	$\bar{1}4$	$\bar{0}\bar{4}$
9	$1\bar{2}$	$1\bar{3}$

Both reports activate only unit 10, whose post-ReLU values are respectively 0.751064 and 1.626101; all other reported values are zero. Replace slot zero's $x_0 \vee \neg x_1$ by $\neg x_0 \vee x_1$. The edited φ_A has sole satisfying valuation 01111; edited φ_B is UNSAT. The checkpoint correctly classifies both originals and both edits, with edited correct-label margins 3.027952 and 6.050618. The original reports are at least 0.251064 from their threshold. Independent-process float32 replay agrees exactly; a float64 diagnostic preserves the classifications and collision. Boolean enumeration is exact, while these native numerical checks are not exact-real verification.

Thus neither the 34-bit report nor even the complete current 32-valuation truth table determines this edit response. Any deterministic decoder of the common report and action fails on at least one formula. This witness is specifically a target-specification obstruction: replacing one clause requires formula structure that current satisfiability semantics do not retain. The bank-level decomposition below separately quantifies additional ambiguity removed by adding the current truth table to the published report. The network contributes a concrete report collision and correct native classifications of the four endpoints; the logical impossibility itself was decidable without it.

The exact lookup audit extends the semantic obstruction to every legal edit in the retained bank. Its target is Boolean-enumerated post-edit UNSAT, stored in `pilot_bank.npz` as the complement of `post_sat`; it is not the transformer's post-edit argmax. Native forwards verify the four witness endpoints and separately specified pilot cohorts, not all 399,360 assessment edit rows. It groups rows by the complete declared decoder input: report, slot, old clause and new clause. A class containing both outcomes has unrestricted worst-case absolute error 1/2; the minimum empirical error instead assigns each class its majority label.

The following table's target is Boolean-enumerated post-edit UNSAT on exactly $1,024 \times 10 \times 39 = 399,360$ assessment rows, not native post-edit argmax.

Assessment report	All groups	Non-singleton groups	Conflicting / non-singleton groups	Rows in conflicts / non-singleton rows	Minimum empirical error
Context only	15,600	15,600	14,375/15,600 (92.147%)	370,803/399,360 (92.849%)	10.553%
Published 34 bits	373,893	23,244	2,474/23,244 (10.644%)	5,375/48,711 (11.034%)	0.622%
Complete current truth table	371,280	25,116	2,077/25,116 (8.270%)	4,543/53,196 (8.540%)	0.526%
Exact stored post-ReLU values	399,360	0	0	0	0%

The all-group fractions are much smaller because singleton groups cannot conflict; the non-singleton denominators expose the relevant finite-bank occupancy. To separate logical target ambiguity from information absent in the published bits, we refine every published-bit/action class with the complete current truth table. Of the 2,474 conflicting published classes, 1,482 contain a conflict even within an identical truth table and 992 conflict only across different truth tables. The published partition makes 2,484 unavoidable empirical errors. The joint published-bit-plus-truth-table partition leaves 1,484 minimum errors. This is the joint floor, not a purely logical target error: truth table alone leaves exactly 2,100 errors, so the published bits remove exactly 616 errors beyond current satisfiability semantics, while truth-table refinement removes 1,000 errors beyond the bits. The latter 1,000, or 0.2504% of all 399,360 edit rows, measures current-semantics information absent from the published bits on this bank. The remaining 1,484 errors require formula structure absent from both the bits and the current truth table; they are not an additional claim about neural prediction failure.

That report-specific component is strongly occupancy- and activity-dependent. Under one deterministic, outcome-blind formula-hash order of the assessment origins, report-specific minimum errors are 3, 55, 172 and 1,000 at nested sizes 128, 256, 512 and 1,024, corresponding to 0.0060%, 0.0551%, 0.0861% and 0.2504% of edit rows. At full size, the 18 origins with zero active published bits account for 272 removed errors (3.8746% of their rows); activity levels one, two, three and four account for 444, 227, 56 and one; levels five through ten account for none. These are post hoc finite-bank sensitivity results, not population estimates. They show why the 1,000/2,484 (40.26%) share of lookup errors removed by supplying the truth table should not be read as a stable estimate of information discarded by the report.

All 1,024 assessment origins have distinct stored post-ReLU vectors, so the final row permits finite-bank lookup memorisation. It supplies neither a compact decoder nor transfer. On all 1,597,440 development edits, published bits have 23,831 conflicting classes among 219,180 non-singleton classes, covering 63,114/524,862 non-singleton rows; refining with the truth table leaves 64.82% of the minimum empirical errors. The truth table alone has 21,170 conflicting classes among 238,797 non-singleton classes, covering 56,879/578,370 non-singleton rows. The archived aggregate counts are reproduced exactly by a zero-fit, zero-forward saved-data audit. These are empirical 0–1 lookup floors; the learned policy results below evaluate fixed learners and do not supply restricted-class optimum certificates.

The earlier action-selection, readiness and four-direction studies are retained in Supplement Appendix I. They supply developmental and exploratory context, with their distinct protocols and limitations, and are not additional protected replications of the all-unit result.

4.4 What the SAT studies establish on their declared cohorts

The SAT studies identify the checkpoint's effective final-MLP state as a graded assignment-aligned report with target-dependent limits. Across all seventeen designated singleton units, every protected threshold-free ranking gate passes, while the frozen combined and threshold-fidelity decisions do not: two units exceed a specificity limit by one case and four fail a positive cell. Errors concentrate near the semantic boundary.

The protected full-space comparison placed the native assignment matches above 63 fixed Haar references. Retrospective analysis changes the explanation and sharpens the like-for-like comparison. The source-nominated 34 units contain effectively all post-ReLU energy and span the numerical-rank-34 state. Within that live span, native assignment matching exceeds every fixed dimension-matched rotation and positive-mixture reference. Structured clause-presence controls reverse direction, and orbit-grouped linear probes predict all assignment labels strongly, including three without equally clean single native axes. Locally fitted 34-component NMF slightly exceeds the native match while recovering 32 native axes at cosine at least 0.99, but incurs 3.15% reconstruction error despite an available exact identity factorisation. This supports assignment alignment and partial recovery of native axes, while leaving prevalence, development and capacity dependence unresolved.

The final readout is affine. All 34 live units add positive SAT evidence, and grouped removals cross more argmax margins than individual removals because that evidence is redundant. Threshold false positives and misses can still contribute useful margin, showing that the fixed affine readout uses magnitudes discarded by the Boolean report. The exact intervention establishes final-cut dependence, not an upstream route. Separately, the 34 threshold bits guide edit choice but cannot determine every legal clause replacement.

Proposed use	Evidence status	Available information	Result and limit
Predict a legal edit from the current published report	Opened-bank exact lookup audit	34 bits, complete truth table and the common action	Joint conditioning leaves 1,484 minimum errors. Truth table removes 1,000 errors beyond bits; bits remove exactly 616 beyond truth alone. Rates are occupancy-dependent.
Select an UNSAT-producing edit	Opened-bank fitted pilot	Initial report or post-ReLU magnitudes and one action menu	Published bits guide choices above action context alone. A one-seed post hoc factorial changes sign with action encoding, so no general bits-versus-magnitudes ordering follows.
Forecast a named unit after a paired edit	Frozen contract on opened states	Fitted readiness bit and assignment-specific action effect	90.95% balanced accuracy; the readiness bit adds 12.24 points over semantic action effect alone. This is an opened-bank predictive result.

Proposed use	Evidence status	Available information	Result and limit
Test all designated singleton-assignment labels	Protected bank and criteria; post hoc diagnostics separated	Seventeen fixed units, five semantic strata and the source threshold	All ranking gates pass. A specificity guard fails for two units and four fail a positive cell. The 36/544 base-cell false positives are Hamming-one SAT negatives.
Characterise the complete final-MLP state	Protected full-space comparison; retrospective geometry, learned-basis and probe controls	All 512 coordinates and the bank-effective 34-dimensional state	Native matching exceeds fixed rotations and mixtures; NMF slightly improves the mean while recovering 32 native axes. Probes recover three assignments lacking equally clean single axes.
Test selected labels on consequential matched edits	Protected four-direction pilot; post hoc causal diagnostics	Fresh matched quartets	Three directions detect at least 31/32 positives at 0.5; unit 411 detects 24/32. Units 374 and 201 are stronger same-assignment axes, so the activity-profile comparisons are not semantic nulls.
Account for the exact final readout	Retrospective exact arithmetic on retained states	Complete state, zeroing, category margins, threshold substitution and refitted readouts	The readout sums positive SAT evidence. All-state removal lowers accuracy to 20.06%; fixed-readout threshold substitution gives 98.15%, while refitted bit and continuous classifiers give 99.12% and 99.64%.

The approximation in the source already allows distribution-dependent errors and weighted assignment margins. The findings delimit its reuse rather than contradict an exact guarantee. The original FTTTT edit witness remains logically unresolved even though unit 189 is a strong FTTTT axis: a coordinate determined solely by the current truth table cannot resolve this witness. The replacement action is already supplied. What is missing is clause-level formula structure, not knowledge of the future action. A clause multiset together with the supplied old and new clauses suffices for Boolean replacement semantics; reproducing sequence-sensitive native responses can additionally require clause order. The stopped negative-score screen remains in the record because its gates were structurally foreclosed, not as evidence against inactive features.

5 Supporting calibration and source audits

Supplement Appendices A–E distinguish missing report information from restricted-decoder failure, using classical minimax/observability results, exact finite certificates and explicit precision repairs on small digit models; these calibrate the audit and are not a new general theory of interpretability. Appendices F–G separately retain the Inception construction failures and Goodfire exported-summary checks, whose numerical and provenance conclusions neither replicate the SAT experiment nor establish a shared upstream mechanism.

6 How to use the audit in interpretability research

A practical investigation should begin with a scientific response question, not a desired representation winner. Specify the intervention cut, side-state channels, output scale, and tolerance. Freeze the explanatory vocabulary and permitted decoder before evaluating the cases used to claim transfer. Count the cost of native and learned encoders on the same ledger.

First measure information loss with identical-report cases. If that lower bound exceeds tolerance, no decoder can rescue the report without more information or a changed target. Next audit the declared decoder. A short contrast rejection identifies a computational limitation even when pairwise collision tests pass. Inspect the positive and negative cases, their feature balances, and their response differences. Their content can guide hypotheses about omitted precision or interactions, but the contrast itself does not name the mechanism.

If a repair menu is available, check whether the entire menu can succeed before launching a long subset search. A full-menu witness can save otherwise wasted optimisation. If it can succeed, exact contrast-cover search can certify menu-relative minimum cost on manageable instances; otherwise use a heuristic with explicit lower/upper gaps. Re-audit after each modification. A feature that breaks one contrast may leave another obstruction intact.

Finally freeze the repaired model and evaluate new inputs, intervention strengths, contexts, and compositions appropriate to the original claim. Keep average risk, worst-case tolerance, and causal selectivity separate. A small predictor error does not establish that the high-level causal graph is correct. Combine this audit with intermediate interventions, causal-abstraction tests, negative controls, and semantic validation.

The precision repair uses native coordinates because they are the available state basis, but its box argument does not privilege that basis. Another invertible or learned basis may make a constrained decoder cheaper or more stable. The comparison becomes informative only when resolution, supervision, transformation cost and decoder capacity are all charged.

7 Implications for alignment research

The decision-relevant lesson is methodological. In this checkpoint, the source 34 are the entire effective final-MLP state on the bank. Native axes are far more assignment-legible than random dimension-matched rotations and fixed positive mixtures, but learned NMF slightly improves the assignment match while largely recovering the native axes. This is evidence for assignment alignment in the observed nonnegative state; the NMF reconstruction and separability qualifications prevent a claim of unique native optimality. This alignment is compatible with both Already Clean and capacity-based superposition accounts. A width sweep spanning dimensions below and above the 32 assignment labels could test capacity dependence, but would discriminate broader explanations only after their distinct predictions were specified.

The exact readout gives a second lesson. The layer is linear at the intervention cut, yet joint removals produce nonlinear decision changes because redundant positive evidence crosses argmax margins. Fixed-readout threshold substitution changes 1.61% of native argmaxes. Under a separate contract, state-only refitted truth classifiers achieve 99.12% from bits and 99.64% from continuous values, a 0.52 percentage-point difference. These are different prediction problems and do not decompose a common loss. For safety work, continuous feature values, readout calibration and decision margins can matter even when axes are semantically legible; any compressed monitor should be fitted and tested end to end on its actual decision target.

The report also separates semantic legibility from action sufficiency. Its bits describe the present SAT computation and guide clause edits, but they do not determine every edited response. Any proposal to use internal features for monitoring or control therefore needs a contract: target, intervention cut, retained state, side information, decoder class, tolerance and evaluation distribution. Claims about deception detection or behavioural certification require direct tests of those targets; this toy does not supply them.

We suggest four checks for native-feature alignment studies, beginning with the effective state actually under comparison:

1. **Effective-state accounting.** Report live energy, rank and the feature-selection channel before interpreting localisation. Compare native axes with alternatives inside the same live subspace and with geometry- and prevalence-matched controls.
2. **Semantic stress test.** Measure sensitivity and specificity on hard negatives, symmetries and compositional cases, preserving continuous scores alongside operational thresholds.
3. **Exact causal accounting.** Separate additive logit effects from nonlinear decision changes. Measure individual, grouped and complement interventions, and distinguish final-cut totals from upstream natural routes.
4. **Action-conditioned sufficiency.** Test the report on the actual monitoring or control decision. Use collision witnesses and restricted-decoder certificates to distinguish missing information from an inadequate reader.

A priority next experiment is a frozen seed-and-width sweep of the same SAT family, with independent unit nomination inside each checkpoint. Widths below, near and above the 32 assignment atoms would reveal whether the observed 34-unit effective state and dedicated native axes persist as operative capacity changes. Each run should report live-unit count, effective rank, native-to-probe gaps, in-span reference comparisons, threshold loss and grouped necessity on both source-distribution and engineered hard-negative banks. Widths below 32 are particularly informative capacity stress tests, but discriminating Already Clean from a capacity-based superposition account requires pre-specified, incompatible predictions. The remaining cells test whether effective width tracks the atom count and whether regularisation, nominal width or seed controls the shared-axis exceptions.

Upstream interchange or path-specific interventions are also needed to determine whether the clean final report is assembled through a stable route. Native units, sparse-autoencoder features and compositional groups should then be compared under the same information, transformation and decoder-cost ledger. The useful question is which representation supports a frozen safety-relevant prediction or intervention, not which looks cleanest in isolation.

8 Conclusion

Cammarata's proposal becomes more precisely testable when "clean" is divided into label ranking, threshold fidelity, effective-state geometry, causal use and action sufficiency. In this checkpoint, published singleton labels rank strongly but fail narrower threshold cells. The source-nominated 34

units form the numerical-rank-34 bank-effective state. Their axes are substantially more assignment-legible than random dimension-matched rotations and fixed positive mixtures, but activation-mass purity is heterogeneous and learned NMF slightly improves matching while recovering most native axes. The defensible description is a graded, assignment-aligned nonnegative state; the local NMF result partly recovers its native axes without establishing a preferred generative representation.

The classifier uses this state continuously. Every live unit adds positive SAT-minus-UNSAT logit margin. Fixed-readout threshold substitution preserves 98.39% of native argmaxes and gives 98.15% task accuracy; refitted bit and continuous classifiers reach 99.12% and 99.64%. For the seventeen-unit group, joint ablation changes more decisions than the sum of individual counts because additive evidence crosses an argmax margin jointly; no interaction term exists at this cut.

The same interpretation has a sharp operational limit: its bits help choose edits but cannot determine every legal edit response. This checkpoint, with nominal width 512 and effective rank 34 on the bank, is consistent with both Already Clean and superposition accounts. The evidence supports direct study of its native mechanism while making no claim about systems with more features than live units. Effective-state accounting, learned-basis comparison, preserved graded information, and upstream interventions are required before the strategy can carry weight in larger alignment settings.

9 Limitations and stopping decisions

A bounded attempt to obtain independently trained competent checkpoints did not remove the single-checkpoint limitation. One development pilot used an incorrect clause grammar and was stopped and excluded. Two subsequent valid-grammar, reduced-regimen pilots reached only 63.10% and 70.475% final validation accuracy after one million presentations each. The latter restored the source's learning-rate warmup but failed its frozen 99% task-competence criterion. Seed and data also changed, so the difference cannot identify a causal warmup effect. No representation or mechanism-bank outcome was inspected, no best checkpoint was selected, and the inexpensive pilot campaign stopped. These failures concern the tested reduced training budgets, not an impossibility of learning the source task or evidence against a mechanistic hypothesis. Appendix L records all three attempts.

The central SAT case concerns one small checkpoint and source-selected units. The early action-selection, negative-score augmentation, readiness and continuous-score studies reuse an opened, initially-SAT development bank. The negative-score protocol and code were frozen before that screen accessed the cached outcomes, but the bank is not protected external evaluation. Its exact policy ceiling is also nearly vacuous for comparing the richer reports: 99.608% of the published report/context classes and every richer class are singletons. The exploratory factorial shows that the bits-versus-magnitudes sign changes with action encoding, but its interaction interval includes zero and the diagnostic uses opened data. The readiness semantic-only baseline, finite-bank information audit, four-unit threshold audit and direction-specific restoration null are likewise later analyses of saved arrays.

The all-unit bank covers all seventeen published singleton-assignment units and the complete native clause grammar, but it remains an engineered equal-weight mixture rather than the source or training distribution. Its 32-case cells make the 0.10 specificity guard discrete: 3/32 passes and 4/32 fails, so the unit-10 and unit-81 decisions rest on one-case margins even though the criterion remains binding. Units share one checkpoint and many formulas; units 55 and 416 use identical base formulas for their common TTTFF label. The formula-orbit bootstrap addresses formula dependence for macro AUROC only and does not sample units, checkpoints or research groups. The local freeze is chronology evidence rather than external preregistration; the static GO review did not run the code, and the complete source-linkage audit occurred after the protected analysis. Historical-ledger exclusion does not prove absence from checkpoint training. The post hoc clause-count regressions,

flipped-variable summaries, ranking-loss decomposition and task-error cross-tab describe stored outcomes and do not inherit protected status.

The full-layer evidence is conditional on the same checkpoint and bank. The source selected J by direct output effect before assigning rules, and J exhausts the effective post-ReLU state on this bank. The training cause of that concentration is not identified. The post hoc source-linkage, energy, live-subspace, positive-mixture, prevalence, probe, purity, NMF, ICA, sparse-dictionary, error-localisation and refitted-readout analyses were designed after opening the protected full-space result. They correct or expose dimensional and geometric confounds but do not acquire protected status. The 63-draw families are constructed references rather than exchangeability nulls. The learned methods were not assigned a protected common complexity budget, and their five seeds do not estimate model-population uncertainty. Standardising each live coordinate before rotation preserves the native mean advantage but does not remove large native zero-ties. The weight audit finds one non-report unit, 386, with nontrivial parameters but no activation on the bank, so the effective-state identity remains bank-conditional. The 34-dimensional result is robust on the unique base formulas, but no bank-sampling interval or checkpoint population is available. Individual and grouped zero ablations are exact at the final affine cut and additive in logits, yet can be off-manifold; within-stratum resampling preserves much of the semantic signal under test. The result establishes a highly legible effective state and exact final-cut use, not prevalence, developmental ancestry, capacity-induced cleanliness, completeness or an upstream causal route.

The preceding matched-edit pilot uses new endpoint queries absent from the earlier native-query ledger, with its design frozen before their responses. That does not supply an external chronology anchor or population coverage. It tests only four of seventeen singleton-labelled units and one variable transposition, and its clause pool excludes types containing $\{x_1, x_2\}$. The later all-unit study resolves those coverage defects for its own engineered bank, but not for natural inputs, other checkpoints or causal use.

The readiness target is a named unit's threshold event, not global SAT action choice. The FTTTT valuation made satisfying by the original witness edit lies outside the four-unit pilot but is represented by unit 189 in the source's wider report. That representation does not resolve the collision: predicting a clause replacement requires formula structure that the current truth table and threshold report discard. Readiness is a deterministic function of the continuous preactivation, and its comparator used different supervision. The source's weighted-margin account remains compatible with the results. The affine pair clamp can be off the natural activation manifold and leaves all other edited-state values fixed. Neither predictive performance nor selective downstream contribution identifies a unique upstream mechanism. The all-unit transposition bank is a dependent consistency check from a permutation-symmetric generator; it is not evidence of learned architectural equivariance.

The supporting digit experiments use one small image dataset. All six seeds share its split, the nonlinear and prospective extensions reuse already inspected images, and automatically selected predicates are not human-written semantic explanations. Interventions of one image are dependent; no writer-independent or model-population claim follows. Finite exact certificates remain valid for adaptively found published instances, but their validity does not estimate prevalence. Full-box precision guarantees can be conservative, and the low-bit repair heuristic has no global optimality or consistent superiority claim. Natural-state pair transfer is deliberately demanding and is not equivalent to single-group attenuation. PCA, random directions and pair-sum reports do not stand in for modern sparse autoencoders, native-group methods or large-model circuits.

The Goodfire additions are audits of exported summaries. They cannot verify hidden selection implementations, recompute AUROC from unavailable per-image responses, reproduce training, or establish developmental ancestry. Inception's twelve illustrations are author-selected calibration, with no verified natural historical cat-leg example. The first deterministic local construction stopped at an inverse-solver residual failure. A separately frozen numerical correction allowed both intended edits to pass the unchanged geometry and pixel-preservation gates, but none of 120 fixed radial-sham candidates met both magnitude-matching requirements. A third, prospectively frozen transverse

family fixed the nominated parent contour pointwise and produced admissible area/RMS matches in both 130-candidate banks. That numerical success did not pass the frozen model-assessor visual gate: two isolated role-blind Anthropic model assessors judged every prepared 224 input ambiguous, and the larger-scale differences admitted seam-thickness, contrast, notch and local-artifact explanations. The route therefore stopped without an edited-image neural forward. This is a construction and model-assessor-gate failure, not evidence for or against the neuron's geometric interpretation or a claim about human perception. It shows why exact outside preservation and global pixel-magnitude matching do not by themselves identify a clean local shape intervention.

The present SAT expansion stops with its bounded positive and negative findings. Further work is justified by a new discriminating question, not by the expectation that additional controls will improve a preferred result. The mathematical ingredients have extensive antecedents. The paper supplies an executable audit, exact calibration and a concrete test of interpretation reuse; it does not establish a general ranking of native neurons, compositional groups or learned sparse features.

Recent identifiability theory sharpens this stopping rule. A representation can be linearly identifiable, axis aligned, or sparsely recoverable under its training data-generating process while remaining untested for a different intervention policy. Conversely, failure of our edit-selection contract does not refute the sufficient conditions or observational conclusions of Zhou and Klindt (2026). The two questions meet only after the latent variables, capacity regime, readout-visible subspace, intervention target and admissible policy context have been connected empirically.

10 Reproducibility and declarations

The private supplied archive completed all 23 replay stages in normal Python: twelve regenerated-JSON comparisons, one native-array bridge and ten supporting checks. All 7,252 native rows were re-executed; retained preactivations, post-ReLU states, residual streams and logits were bitwise identical. The runner deliberately rejects optimized Python because some historical checks require assertions. Separate assertion-independent headline and fresh-bank recounts passed in normal Python and actual `-O` mode; this is not an optimized replay of all 23 stages.

The public derivative has a narrower redistribution scope: 18 runnable stages and five explicitly historical, nondistributed image-dependent stages, because underlying photograph rights have not been established. Those omissions cover the Inception saved-data, construct-validity, geometry-failure, geometry-bisection and fixed-seam checks; they do not remove an adverse outcome from the narrative. The private 23-stage result must not be read as a clean replay receipt for the public derivative. The public manifest and stage inventory identify its own exact coverage. No public link to the private supplied archive is implied.

The versioned public evidence ZIP supplies the checkpoint, distributable retained data, frozen protocols, corrected analysis code, applicable source licences and a portable `reproduce.py` entry point. It does not redistribute the inherited upstream `model.py`, `plot.py` or `modular_addition_utils.py`: an applicable redistribution grant for their original upstream source has not been established. Before native replay, readers must retrieve the specified public Palumbo source files separately and pass their directory through `reproduce.py --upstream-directory`; the launcher verifies their pinned hashes and stages them into its disposable replay workspace. The public package is therefore not a self-contained distribution of all model code. Its own manifest must be verified before and after replay from a clean extraction; final public-export validation is recorded separately. The native-array bridge re-executes the 7,252 protected inputs; protected statistics can then use the archived arrays through the explicitly checked equivalence. Supporting saved-data checks are not full model re-executions. The historical 23-stage counts above describe the supplied private input, not the reduced public stage inventory.

Original freeze hashes are fixed in the launcher. Each top-level replay stage activates a Python audit hook that rejects original-workspace access and self-tests that rejection; native-library I/O

may bypass Python audit events, and nested supporting subprocesses do not automatically inherit this hook. Regenerated absolute provenance paths must resolve to existing files inside the disposable replay workspace and match the full archived relative suffix; changed fields are logged. This is an accidental-read safeguard, not an operating-system filesystem sandbox or a second-host reproduction. The three runtime inventories, pinned package snapshots and stage-to-environment map accompany the release. Analysis uses Python 3.14.7/NumPy 2.4.2; native execution and readout-accounting v4 use Python 3.11.14/NumPy 1.26.4/Torch 2.1.2; corrected v6 and NMF diagnostics use Python 3.11.14/NumPy 2.4.6/SciPy 1.17.1/scikit-learn 1.9.0. Earlier diagnostic exports remain explicitly superseded. A machine-readable claim map connects reported values to artifact keys and replay stages. Historical absolute paths in frozen records are provenance, not required input locations.

The clean replay does not retrain models, repeat legacy optimiser fits, rerun early SAT witness/development/readiness/pilot forwards, execute the Inception TensorFlow graph or repeat Anthropic assessor calls. Those results retain their separately described saved-data evidence. Supplement Appendix J distinguishes original execution, historical checks and this release replay. SHA-256 manifests identify the release; no persistent repository DOI has yet been assigned.

Data availability. Retained study materials accompany the manuscript and evidence ZIP. The pinned Palumbo repository carries an MIT licence, retained byte-for-byte; the licence file was verified against commit `35d1e4105265ab0ca74bc7daf9371fc8f82bba81`. Dataset and other third-party source terms remain applicable. No blanket replacement licence is imposed on those materials.

Ethics. No new human participants were recruited. The study uses an existing digit-image dataset, generated Boolean formulas, synthetic hidden-state interventions and public source illustrations. No deployment or consequential individual decision is made.

Attribution and contributions. Scholarly attribution is Anonymous. The project maintainer and publisher administered the release; that role is not evidence of human research authorship, independent proof validation or peer review. Methods, derivations, software, analysis, computational checking, visualisation and writing received AI assistance as detailed below. The named systems are disclosed as research tools, not listed as human authors.

AI disclosure. Retained Codex session metadata records GPT-5.6-sol and GPT-6-astra agents, used for literature discovery, mathematical derivation, software, analysis, manuscript preparation and review prompts. Claude Opus 5.5 supplied static protocol and full manuscript criticism. Separate role-blind Claude Opus 5.5 and Sonnet 5.5 calls supplied the two visual-assessor judgements in the stopped image study. These Anthropic calls used adaptive high effort and a 128,000-token output ceiling and completed naturally; exact per-call settings, inputs and outputs are retained. Mathematical derivations were AI-assisted and model-reviewed; finite certificates were checked computationally with the stated exact arithmetic. No human proof validation, proof-assistant formalisation or independent human peer review is claimed. Model criticism and assessor outputs share model-related limitations.

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